

TARGET-90

JKBOSE HUMANITIES CLASS- 11TH

WORK, ENERGY AND POWER

One Shot

PHYSICS

By-Wajahid Sir (Gabbar Sir)





Topics to be Covered

A Work ✓

B Types of Work ✓

C Positive Work ✓

D Negative Work ✓

E Energy ✓



Topics to be Covered

F

Type of Energy ✓

G

Kinetic Energy ✓

H

Kinetic Energy (Work Energy Theorem) ✓

I

Potential Energy ✓

J

Law of Conservation of Energy ✓



Topics to be Covered

K Work done By spring ✓

L Spring Potential Energy ✓

M .. Collision

N .. inelastic and inelastic collision

Work, Energy and Power

Work done by a constant force and a variable force; kinetic energy, work energy, Theorem, power. Notion of potential energy, potential energy of a spring, conservative forces; non-conservative forces, motion in a vertical circle; elastic and inelastic collisions in one and two dimensions.





WORK

1



Scalar Quantity

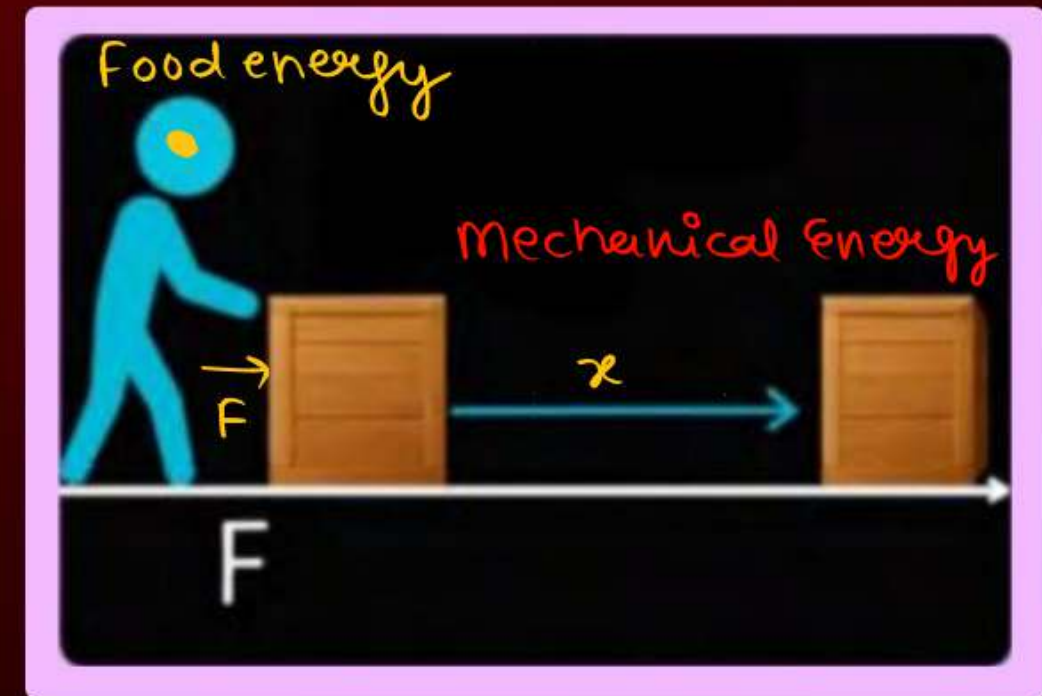
★ Physics main jo work hota hai, wo society K work se bilkul alag hota hai. ❄ ❄

★ Work is mode of transfer of energy.

★ Work done is the dot product of force and displacement

$$W = \vec{F} \cdot \vec{x}$$

$$W = Fx \cos \theta$$



$$W = \vec{F} \cdot \vec{x}$$

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$W = Fx \cos \theta$$

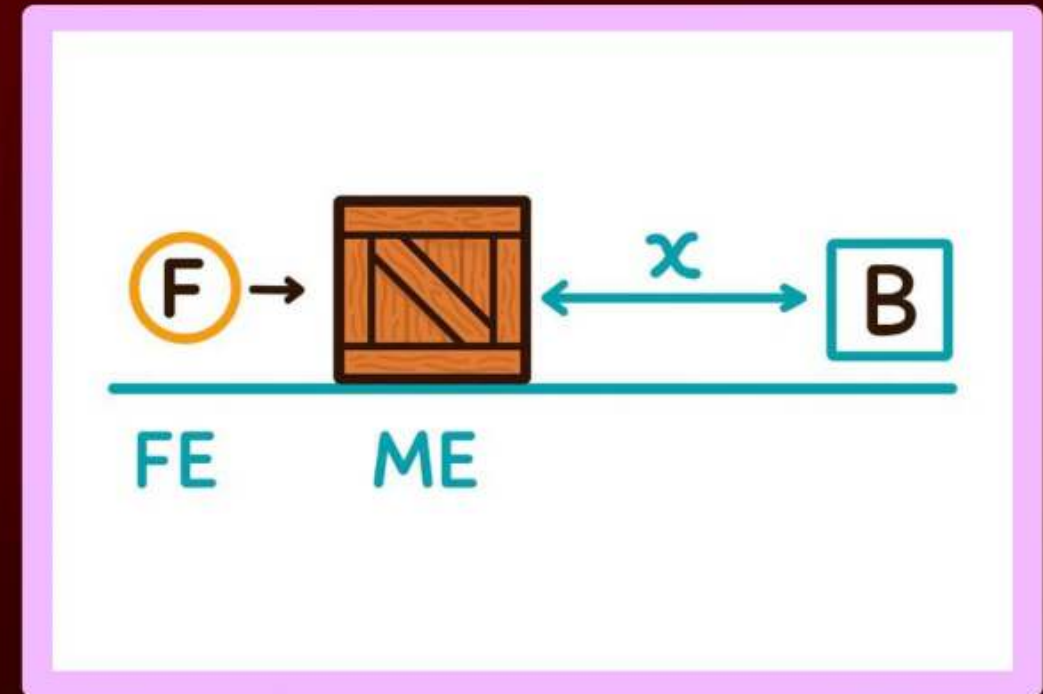
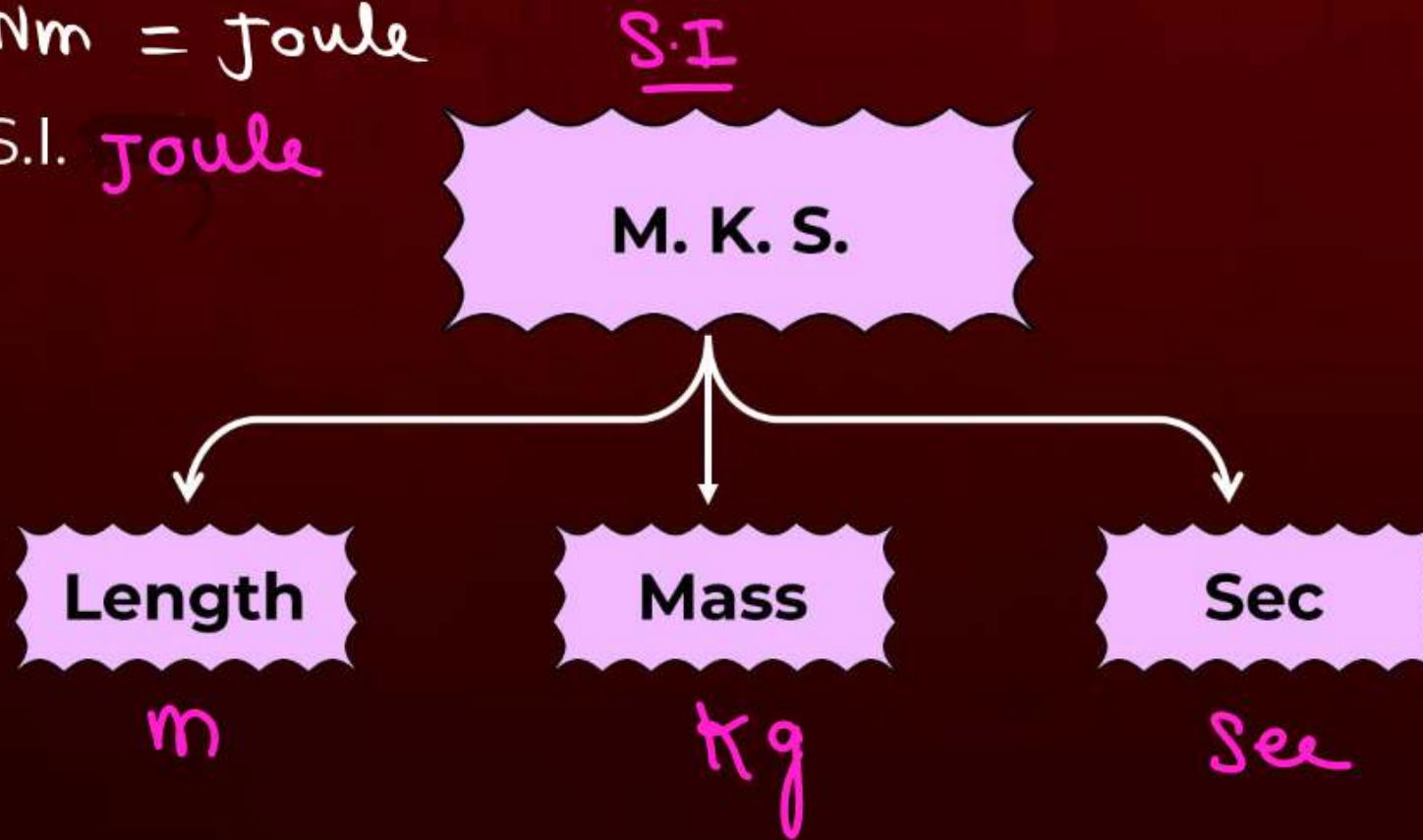


WORK

$$\cos \theta = \frac{B}{H} = \frac{\cancel{xx}}{\cancel{xx}} = 1$$

★ $\underline{W} = \underline{F} \underline{x} \underline{\cos \theta}$
 $= Nm \times 1$
 $= Nm = \text{joule}$

★ Unit: S.I. **joule**





WORK

C. G. S.

Length

cm

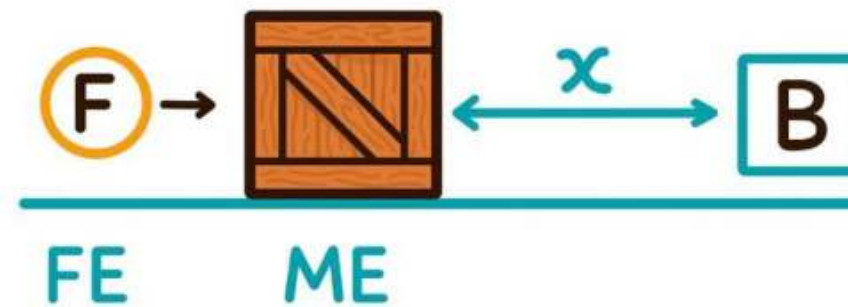
Mass

gm

Sec

Sec

- $W = F x \cos \theta$
- $W = \text{dynes cm}$
- $W = \text{erg}$



"Work = Scalar"



WORK

S.I $\rightarrow$ C.G.S

★ $1\text{ m} = 100\text{ cm}$

★ $1\text{ kg} = 1000\text{ gm}$

★ $1\text{ J} = 10^7\text{ erg}$

★ $1\text{ J} = 10^7\text{ erg}$

★ $1\text{ N} = 10^5\text{ dynes.}$

Imp





TYPE OF WORK

Work

Depending on Angle θ b/w F and x

Positive

Negative

Zero



① POSITIVE WORK

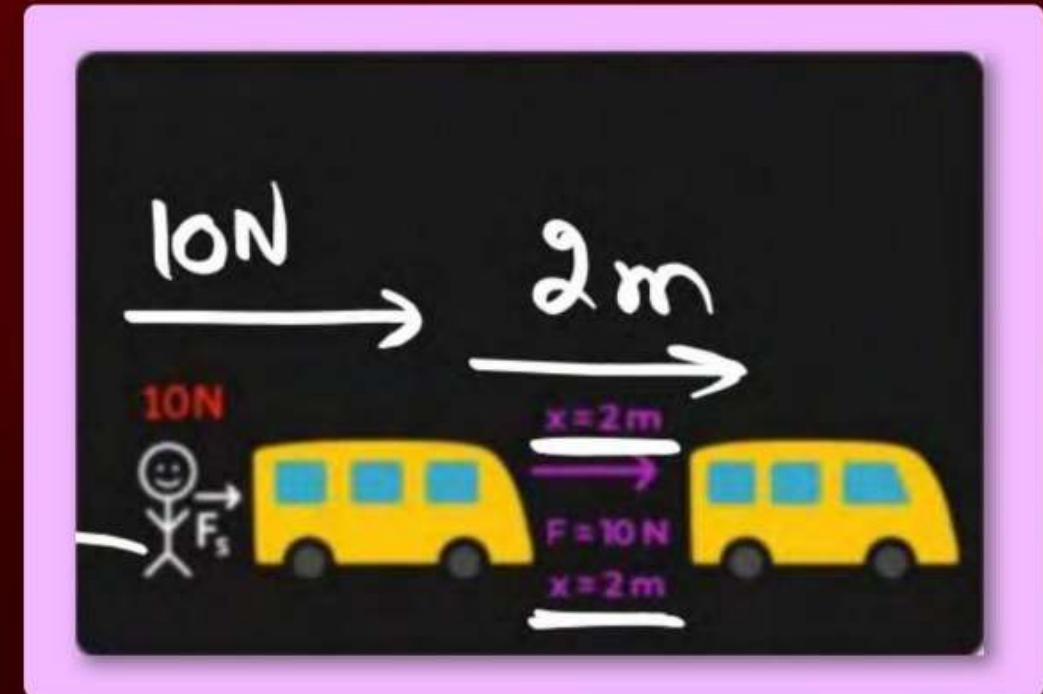
- The positive is said to be done when force and displacement Both are in same direct

★ $W_s = Fx \cos \theta$ $\begin{matrix} \longrightarrow F \\ \longrightarrow x \end{matrix} \theta = 0$

$$W_s = F_s x \cos \theta$$

$$W_s = 10 \times 2 \cos 0^\circ = 1$$

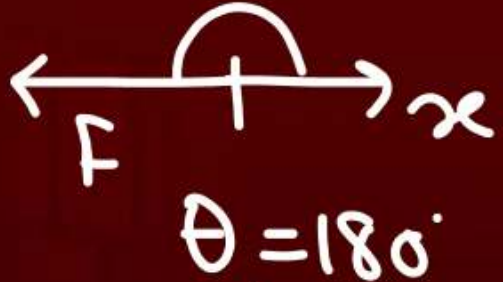
$$W_s = +20 \text{ joule.}$$





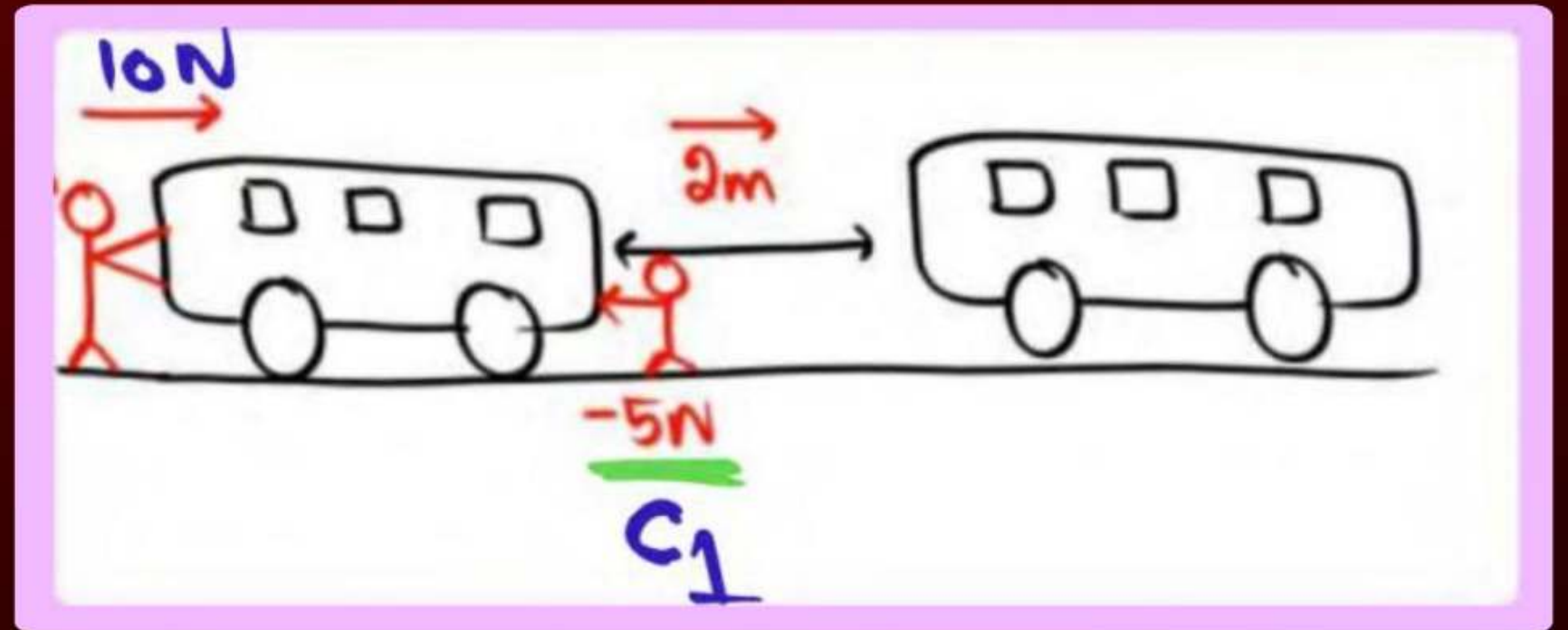
NEGATIVE WORK

- When the force and displacement Both are in opposite direction

✦ $W_{c_1} = \underline{F} \underline{x} \underline{\cos \theta}$ 

$W_{c_1} = 5 \times 2 \cos 180^\circ$

$W_{c_1} = 10 \times -1$
 $= -10 \text{ Joule}$





③ ZERO WORK

- when the Force and displacement
Both are perpendicular to each other

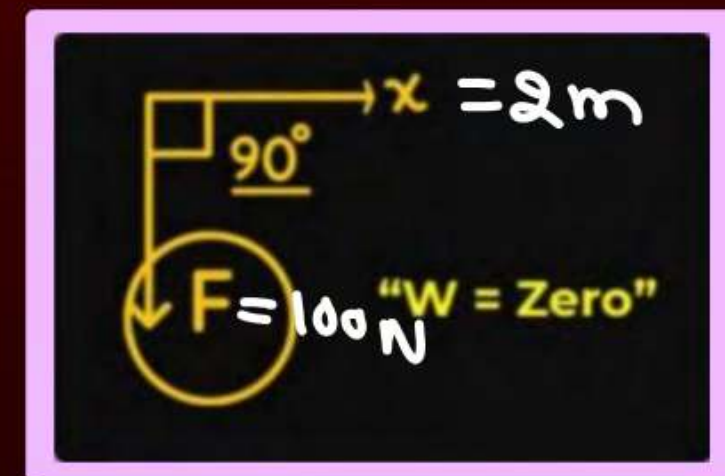
✦ Work done by C_2

$$W_{C_2} = F_{C_2} x \cos \theta \quad \theta = 90^\circ$$

$$= 100 \times 2 \times \cos 90^\circ$$

$$= 200 \times 0$$

$$= 0 \text{ J}$$





WORK

Work

depend on Force

Work done by
constt Force

Force = 2N = Constant

(Jo na badle)

Work done by
Variable Force

Force = Change (variable)
Work done by Spring Force.

- * The work is done by constant force in which value of force does not change

(•) लालचोके

10N $\rightarrow$ m A

10N $\rightarrow$ m B

10N $\rightarrow$ m C

10km

$\theta = 0^\circ$

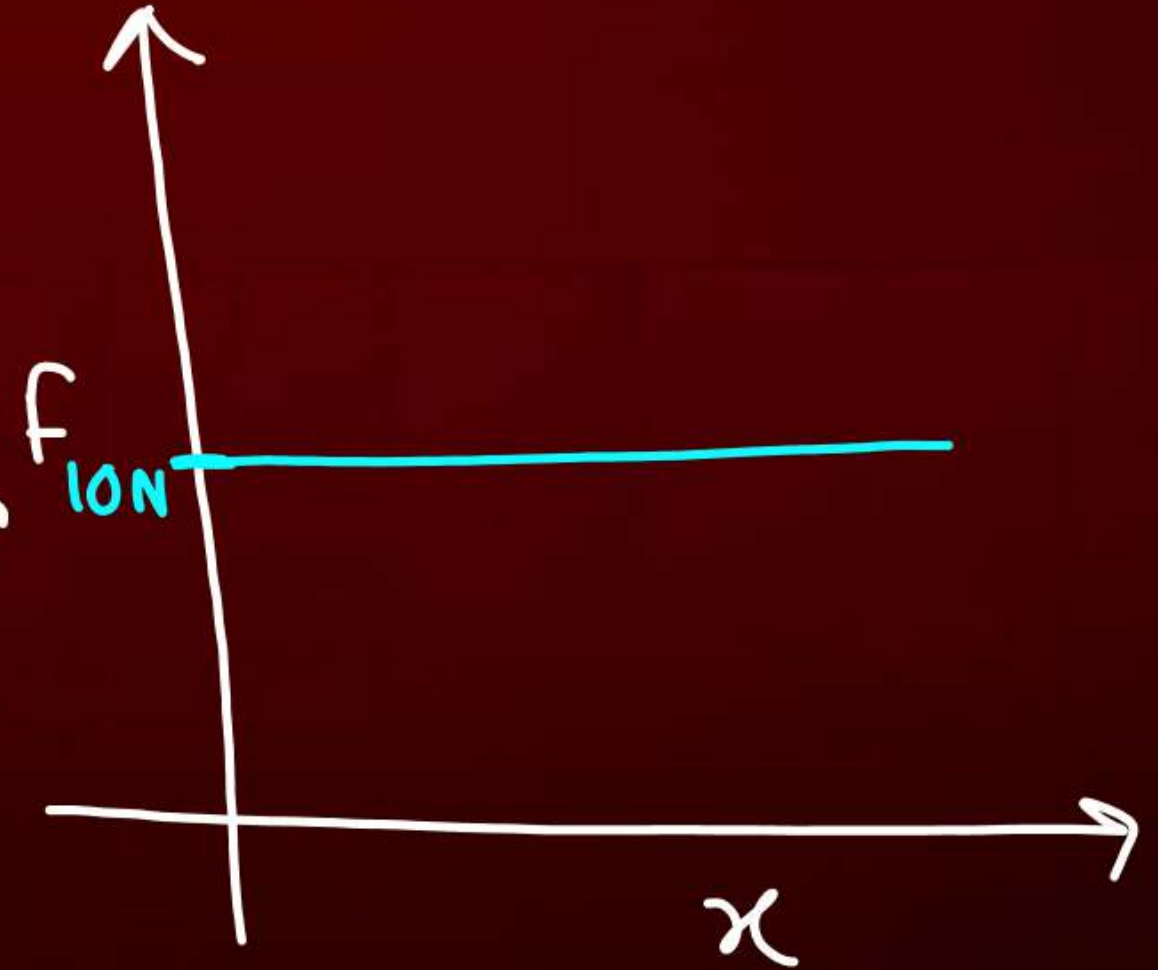
Raj bagh

Home

$$W = Fx \cos \theta$$

$$W = 10 \times 3000 \text{ m} \cos 0^\circ$$

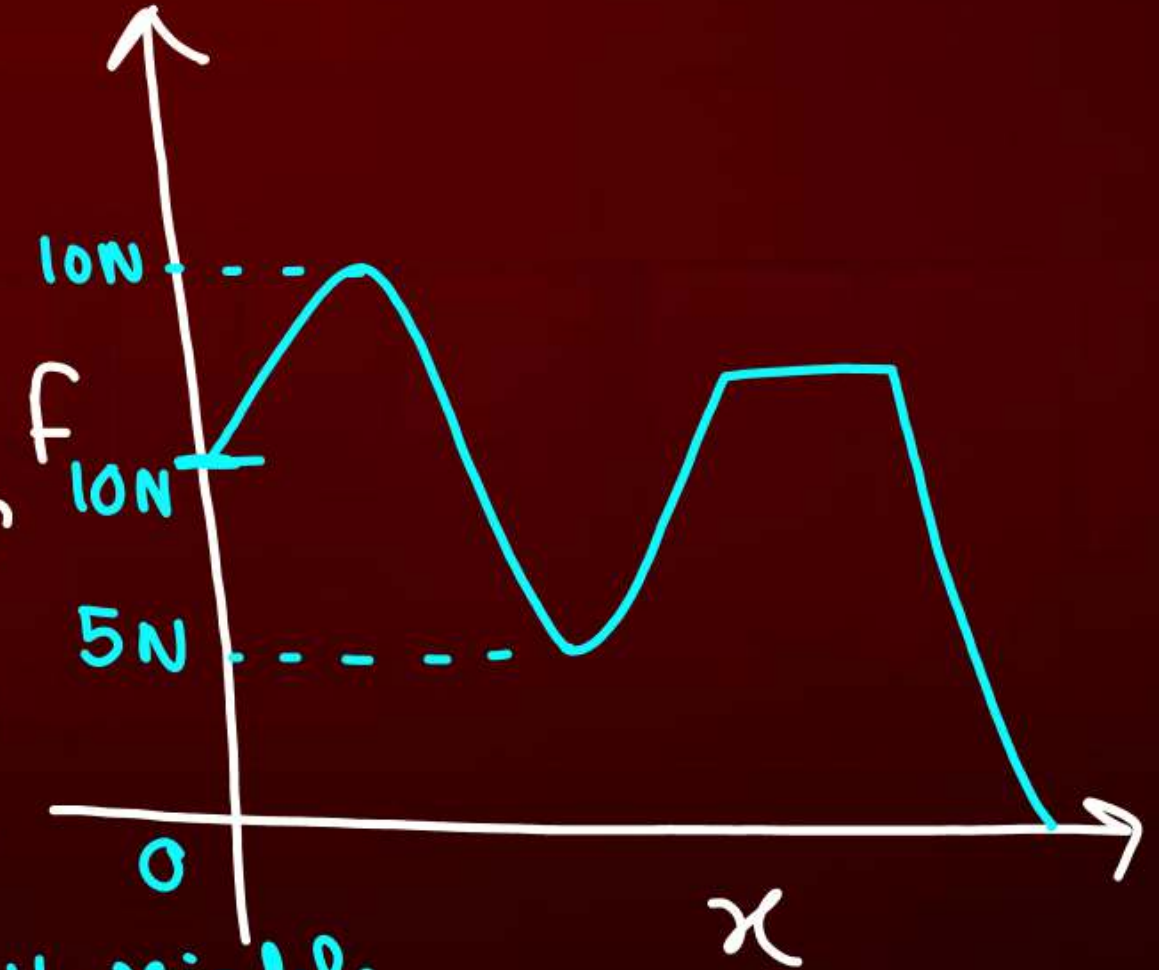
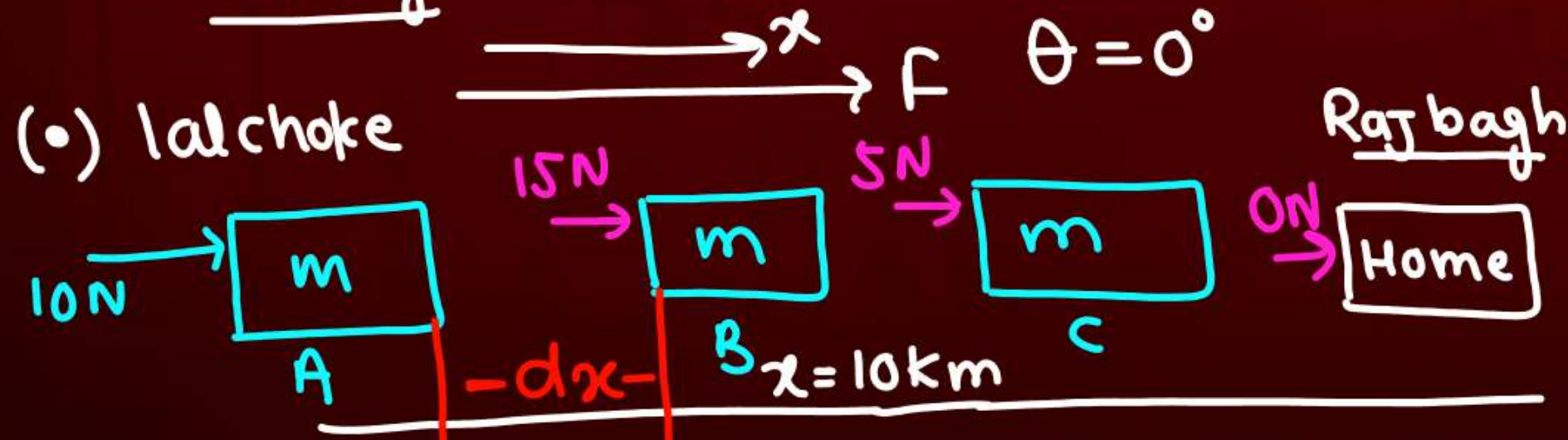
$$W = 30,000 \text{ Joule}$$





① Work Done By ^{Variable} ~~Const~~ Force

* The work is done by changing force in which value of force Chang



$$dw = \vec{F} \cdot d\vec{x}$$

$$\int dw = \int \vec{F} \cdot d\vec{x}$$

$$\text{Work} = \int \vec{F} \cdot d\vec{x}$$

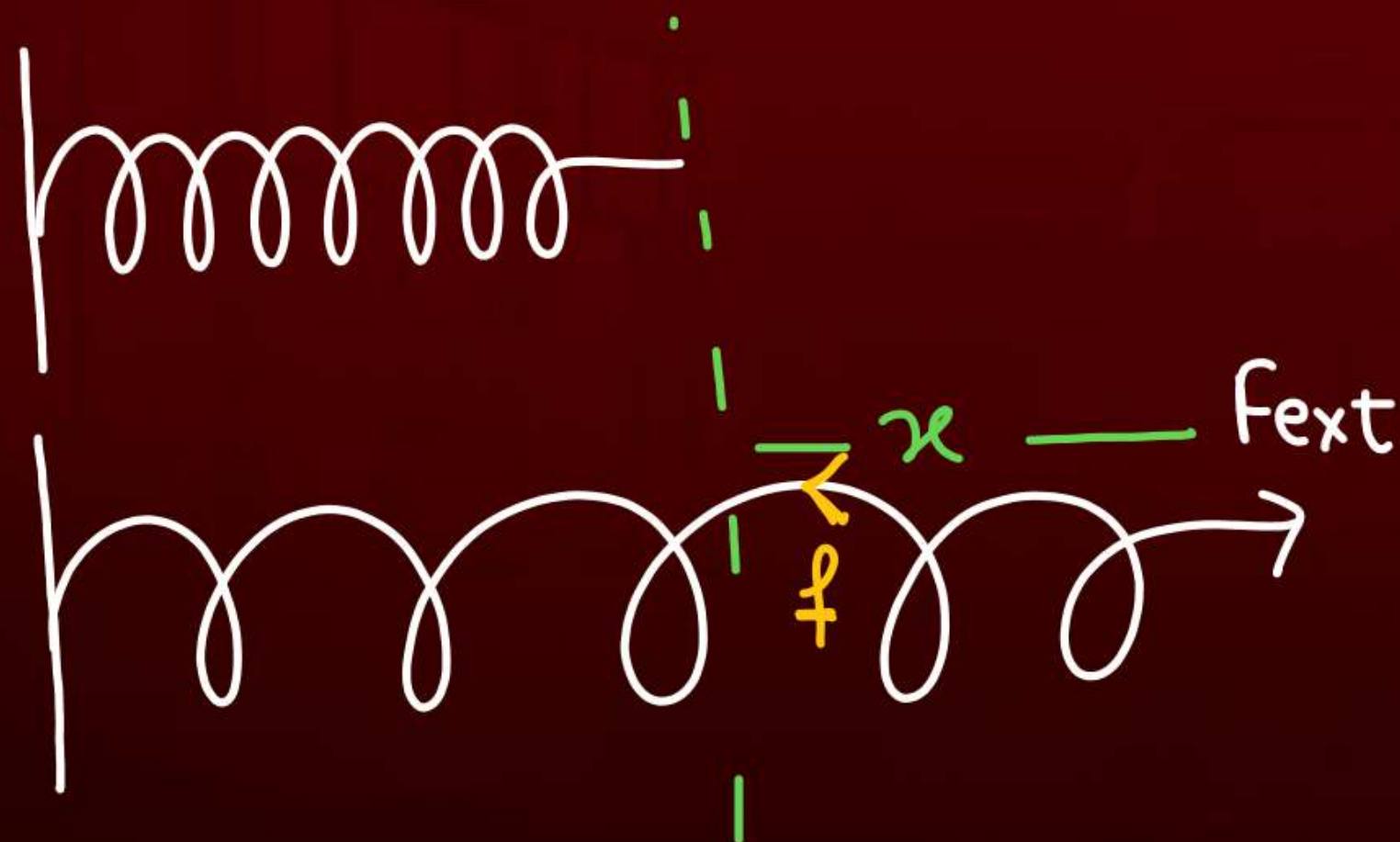
When force is variable

$$W = \int \vec{F} \cdot d\vec{x}$$



Spring

- Spring is made up of Steel



Spring force

2mark



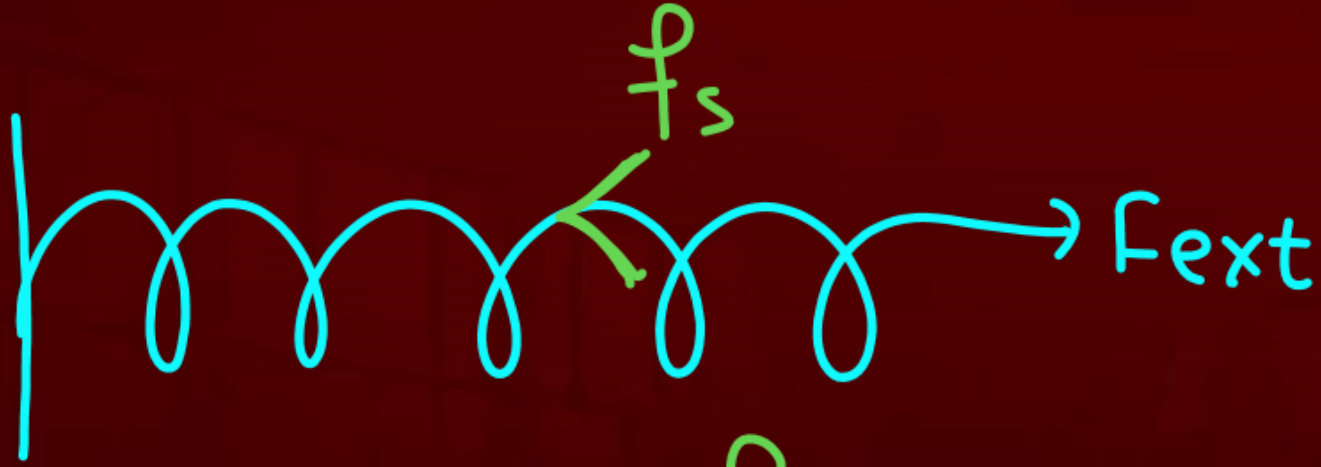
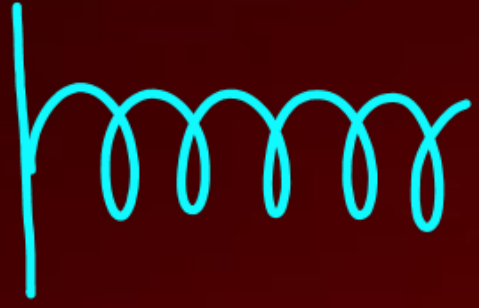
The Restoring force developed in Spring when it is Stretched.

$$\uparrow F_s \propto x \uparrow$$

$$F_s = -Kx$$

$K \rightarrow$ Spring Constant.

Variable force



$$f_s = -kx$$

$$\frac{f_s}{-x} = \textcircled{k \uparrow}$$

Spring constant
(Spring kitni ziddi hai)

$$\frac{f_s}{x} = k \downarrow$$



Work Done By Spring Force

(•) let us consider the spring of Spring Constant K and it is stretched through a displacement (x)

2025

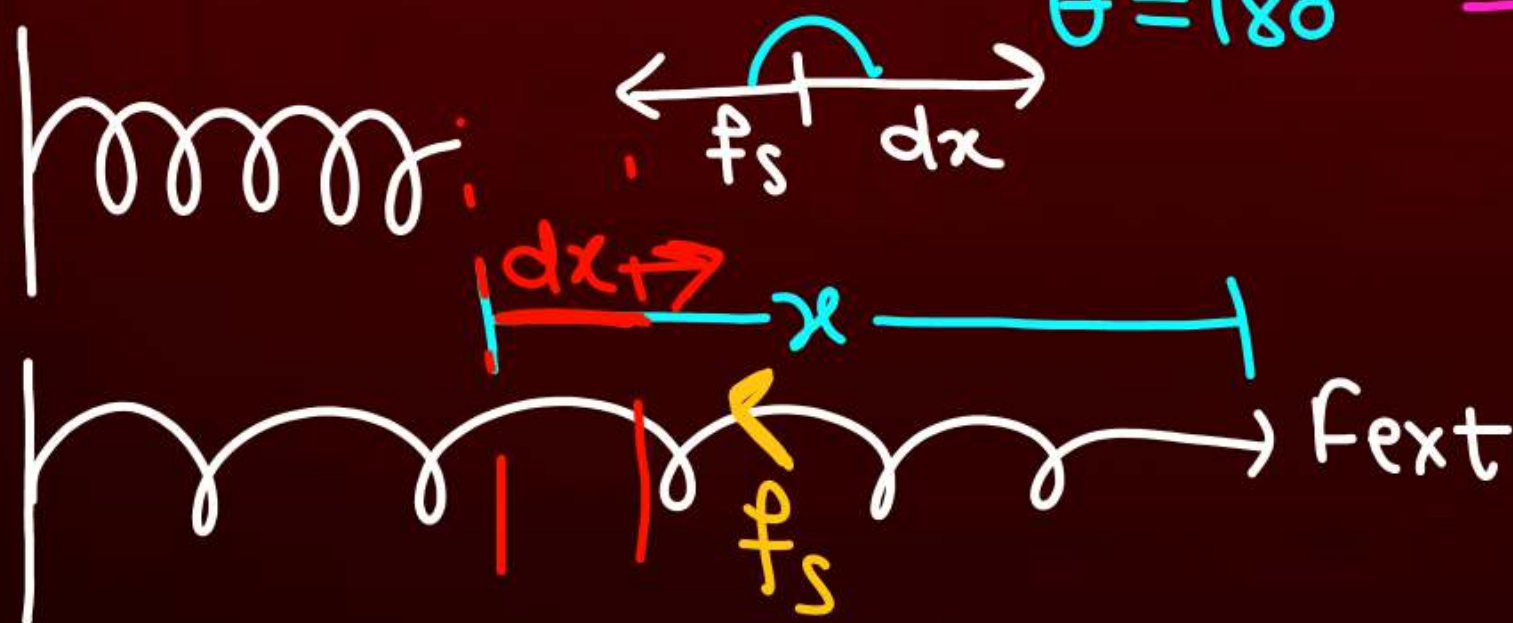
2024

2023

2021

2016

$\theta = 180^\circ$



$$\# \vec{f}_s = -kx \rightarrow \text{Variable}$$

$$dw = \vec{f}_s \cdot d\vec{x}$$

$$dw = f_s dx \cos \theta \quad f_s = -kx$$

$$dw = f_s dx \cos 180^\circ \rightarrow -1$$

$$dw = f_s dx \times -1$$

$$dw = -f_s dx$$

$$f_s = kx$$

$$dw = -kx dx$$

on integrating b/s

$$\int dw = \int -kx dx$$

$$\underline{\int dw} = \int \underline{-kx} dx$$

$$\text{WORK} = -k \int x' dx$$

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

$$W = -k \frac{x^{1+1}}{1+1}$$

$$W = -\frac{kx^2}{2}$$

$$W = -\frac{1}{2} kx^2$$



Energy

- ✦ The Capacity of do work is ~~Kin~~ ^{called} Energy
Papa Beta
Energy → Work
~~~~~      ~~~~~
- ✦ S.I. unit of Energy is Joule
- ✦ C.G.S. unit of Energy is erg
- ✦ Energy is a Scalar quantity



## TYPE OF ENERGY

**Solar Energy**



**Wind Energy**



**Hydropower Energy**



**Petroleum Energy**



**Coal Energy**



**Tidal Energy**





## SOLAR ENERGY

- ✦ The energy you get from the sunlight.





## WIND ENERGY

- ✦ The energy you get from the wind/air.





## HYDROPOWER ENERGY

- ✦ The energy you get from flowing water.





# Petrol ENERGY

- ✦ The energy you get from petrol.





## COAL ENERGY

- ✦ The energy you get from coal.





## TIDAL ENERGY

- ✦ The energy you get from waves of the ocean.





## TYPE OF ENERGY

**Food Energy**

**Sound Energy**

**Heat Energy**

**Electrical  
Energy**

**Light Energy**



# MECHANICAL ENERGY



Energy

The energy related  
to  
motion/Kinematics

Mechanical  
Energy

Kinetic Energy  
(K.E.)

$$K.E. = \frac{1}{2}mv^2$$

Potential Energy  
(P.E.)

Example

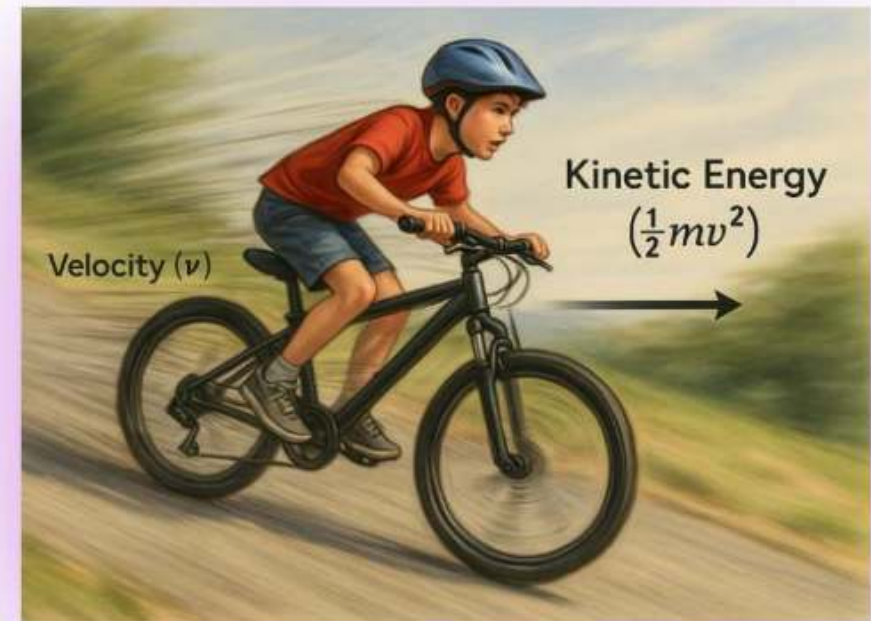
$$P.E. = mgh$$



## KINETIC ENERGY

✦ The Energy by virtue of motion of  
the particle / substance

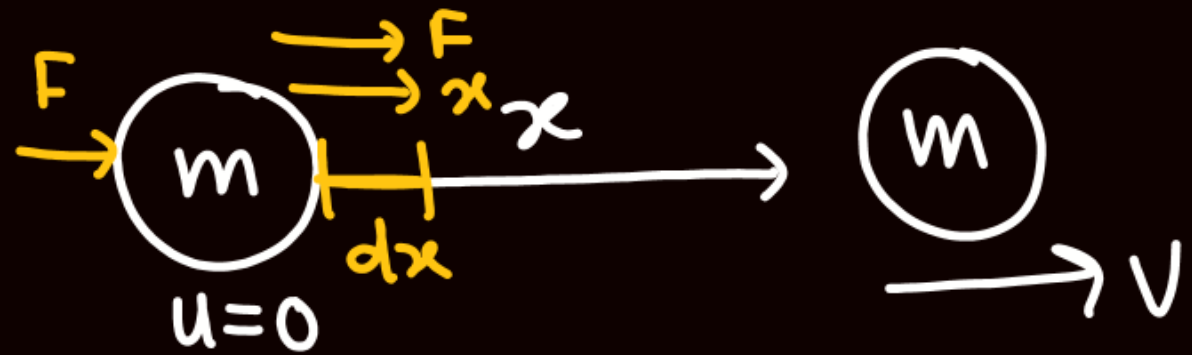
$$K.E. = \frac{1}{2}mv^2$$



# Derivation of Kinetic Energy

• let us consider the body of mass (m) at Rest

⇒ we will master Apply the force (F) on it and



It displaced through a displacement ( $x$ ) and attain the velocity ( $v$ )

$$dw = F dx \quad F = ma$$

$$dw = m a dx \quad a = \frac{dv}{dt}$$

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

$$dw = m v \frac{dv}{dx} dx$$

$$a = \frac{dv}{dt} \times \frac{dx}{dx}$$

$$dw = m v dv$$

on integrating b/s

$$\int dw = \int m v dv$$

$$W = m \int u^u v^v dv$$

$$W = m \left[ \frac{v^{1+1}}{1+1} \right]_u^v$$

$$a = \frac{dv}{dx} \times \left( \frac{dx}{dt} \right)^v$$

$$a = \frac{dv}{dx} v$$

$$a = v \frac{dv}{dx}$$

$$\underline{W} = m \left[ \frac{v^2}{2} \right]_u^v$$

$$K \cdot \varepsilon = m \left[ \frac{v^2}{2} - \frac{u^2}{2} \right]_0^v$$

$$K \cdot \varepsilon = m \left[ \frac{v^2}{2} - \frac{0}{2} \right]$$

$$K \cdot \varepsilon = \frac{mv^2}{2}$$

$$K \cdot \varepsilon = \underline{\frac{1}{2}mv^2}$$



## WORK ENERGY THEORM

2026 2023 2016  
2025 2022  
2024 2021



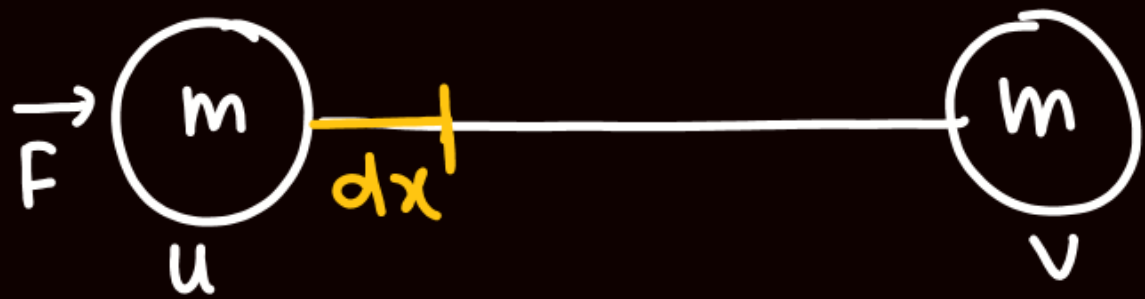
# Work is the change in  
K.E

$$W = \Delta K.E$$

Proof

## Proof

- let us consider the mass ( $m$ ) with initial velocity ( $u$ )



- let  $F$  be the force acting on that ( $m$ ) and body displaced through the displacement ( $x$ ) and attain the final velocity ( $v$ )

$$dW = \underline{F} dx \quad F = ma$$

$$dW = m \underline{a} dx \quad a = v \frac{dv}{dx}$$

$$dW = m v \frac{dv}{dx} dx$$

$$dW = m v dv$$

on integrating b/s

$$\int dW = \int_u^v m v dv$$

$$W = m \int_u^v v dv$$

$$W = m \left[ \frac{v^2}{2} \right]_u^v$$

$$W = m \left[ \frac{v^2}{2} - \frac{u^2}{2} \right]$$

$$W = m \frac{v^2}{2} - m \frac{u^2}{2}$$

$$W = \frac{1}{2} m v^2 - \frac{1}{2} m u^2$$

$$W = K_f - K_i$$

$$W = \Delta K \cdot \xi$$

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$



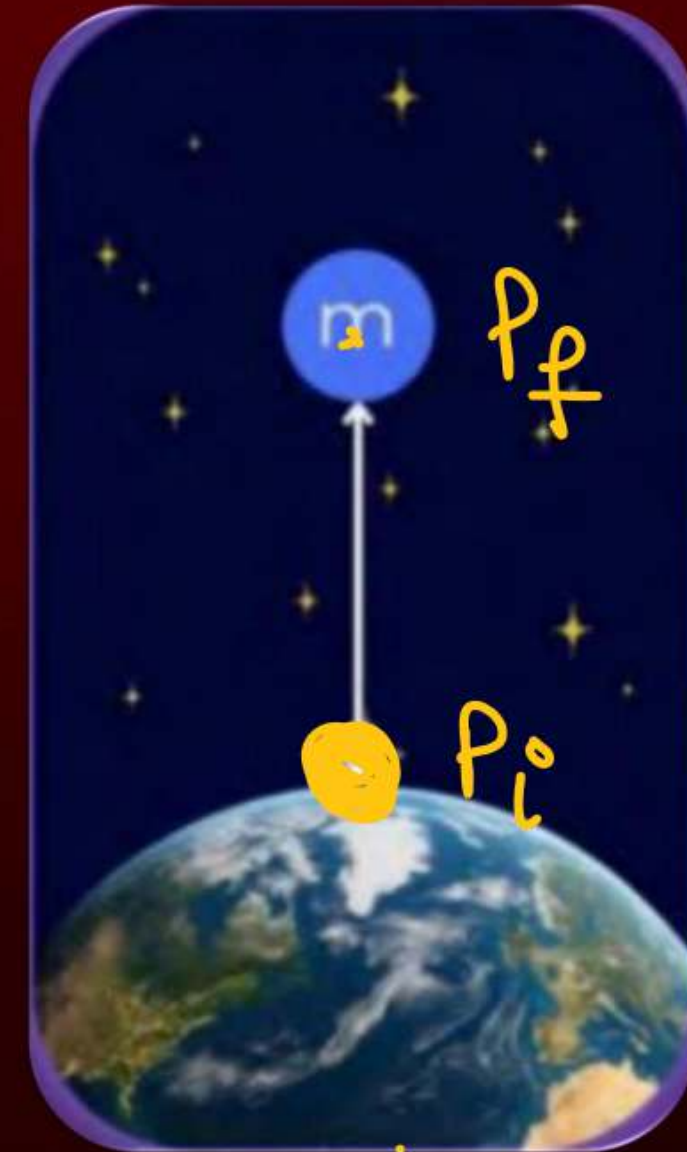
# POTENTIAL ENERGY

✦ The energy due to Zabardasti.

Or

✦ ✓ The P.E. due to the position of the particle.

$$\text{P.E.} = -W_{\text{internal force.}}$$

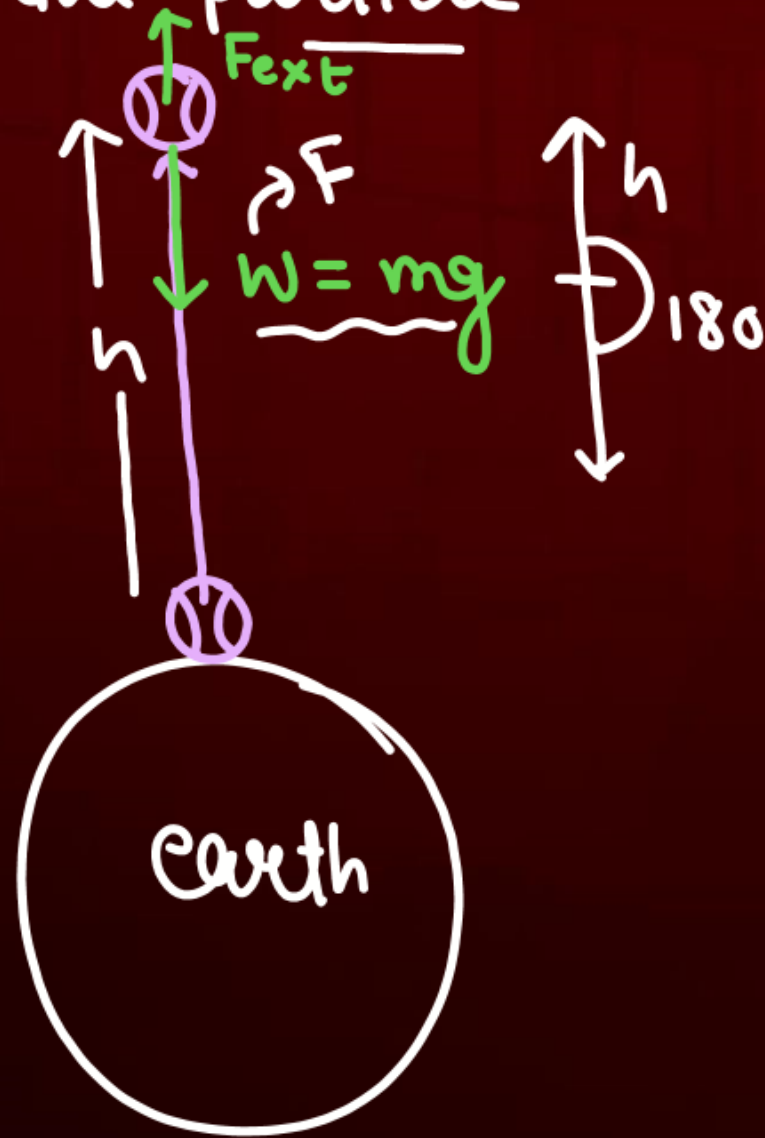


$$\underline{P.E} = -W_{\text{Internal force}}$$

$$U = -W_I$$

## (I) Gravitational P.E

- The P.E due to the Relative position of the particle



- $W_I = \text{work due to gravitation force}$   
( $W = mg$ )

- $W_I = Fx \cos \theta$

$$W_I = \underline{W} \underline{h} \underline{\cos 180} \rightarrow -1$$

$$W_I = mgh (-1)$$

$$W_I = -mgh$$

$$U / \text{P.E} = - \underline{W_I}$$
$$= -(-mgh)$$

$$U / \text{P.E} = +mgh$$



## GRAVITATION P.E

$$P.E = mgh.$$

★ Where,

P.E. = Potential Energy

$m$  = mass of body

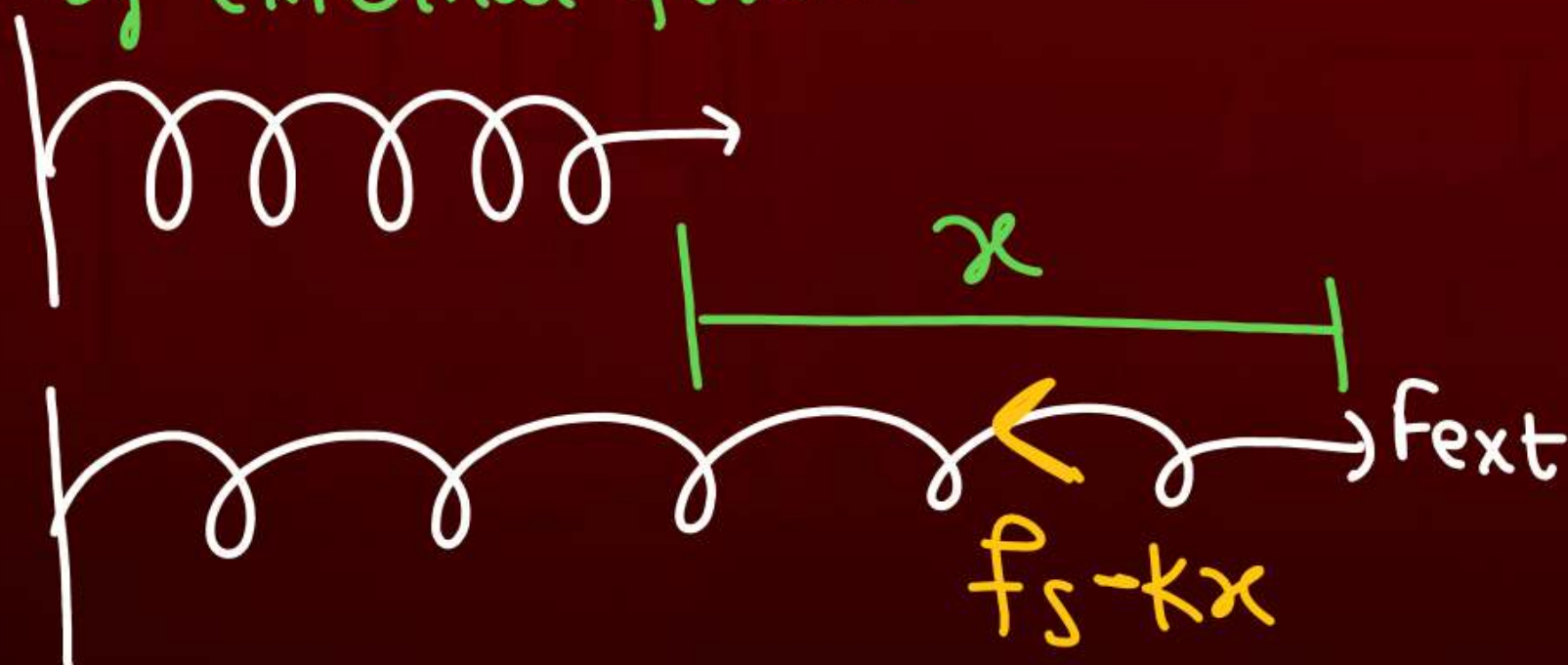
$g$  = Acceleration due to gravity

$h$  = height



## ② PE Store in the spring

- when you stretch the spring by external force.



# The P.E Store in the Spring

$$U = -\underline{W_I}$$

The work done by internal force (Spring) force



## Work Done By Spring Force

(•) let us consider the spring of Spring Constant  $K$  and it is stretched through a displacement ( $x$ )

2025

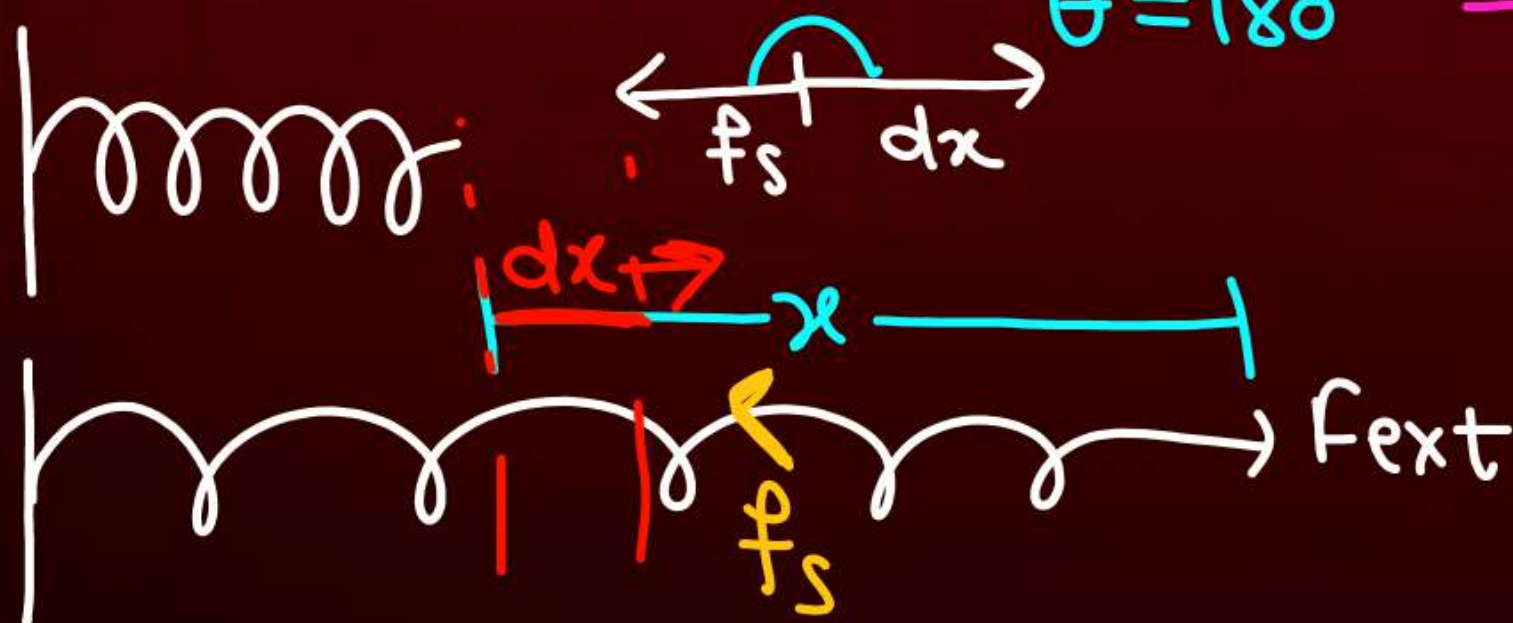
2024

2023

2021

2016

$\theta = 180^\circ$



$$\# \vec{f}_s = -kx \rightarrow \text{Variable}$$

$$dw = \vec{f}_s \cdot d\vec{x}$$

$$dw = f_s dx \cos \theta \quad f_s = -kx$$

$$dw = f_s dx \cos 180^\circ \rightarrow -1$$

$$dw = f_s dx \times -1$$

$$dw = -f_s dx$$

$$f_s = kx$$

$$dw = -kx dx$$

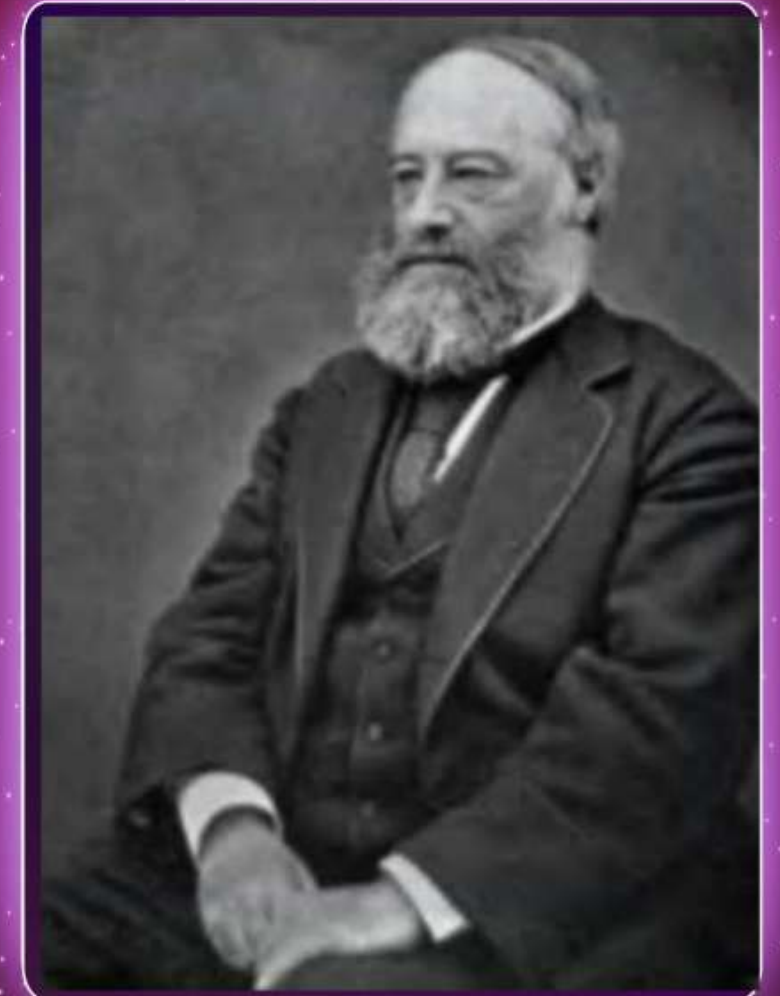
on integrating b/s

$$\int dw = \int -kx dx$$



## LAW OF CONSERVATION OF ENERGY

- ✦ James Prescott Joule was an outstanding British physicist. He is best known for his research in electricity and thermodynamics. Amongst other things, he formulated a law for the heating effect of electric current. He also verified experimentally the law of conservation of energy and discovered the value of the mechanical equivalent of heat. (The unit of energy and work called joule, is named after him)



$$\underline{\int dw} = \int \underline{-kx} dx$$

$$\text{WORK} = -k \int x' dx$$

$$W = -k \frac{x^{1+1}}{1+1}$$

$$W = -k \frac{x^2}{2}$$

$$W = -\frac{1}{2} kx^2$$

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

$$W_I = -\frac{1}{2} kx^2$$

$$U = -W_I$$

$$U = - - \frac{1}{2} kx^2$$

$$U = +\frac{1}{2} kx^2$$

$$\text{Mechanical Energy} = K.E + P.E$$



## LAW OF CONSERVATION OF ENERGY

2024  
2021



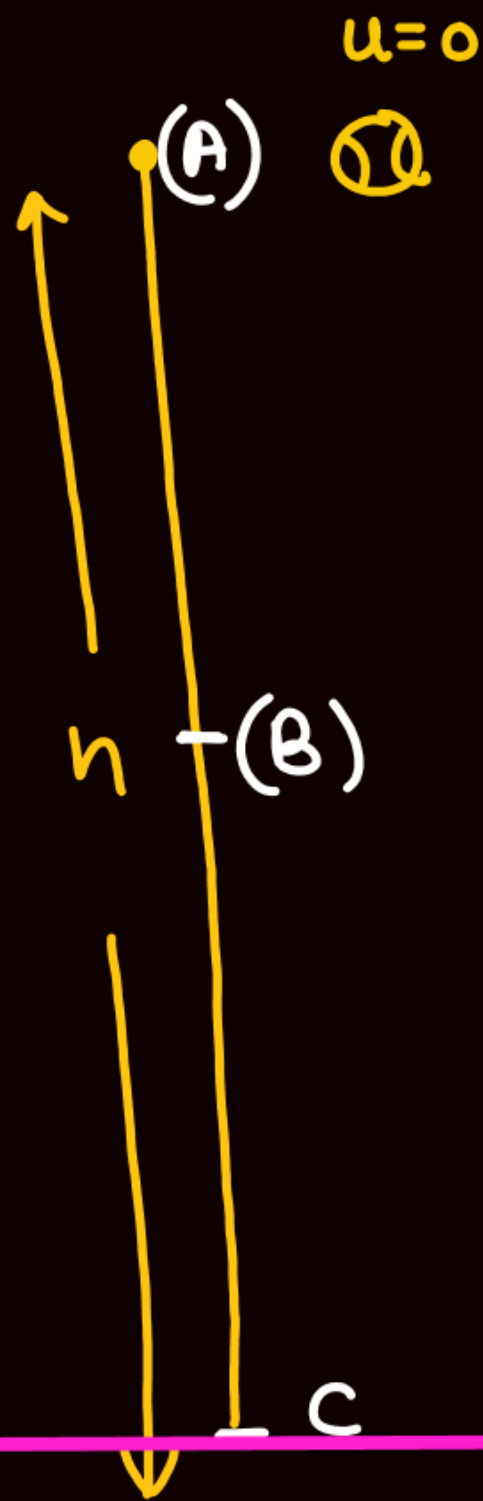
- ✦ Energy can never be created nor be destroyed but can be transferred from one form to another form, total energy of the system remain constant.

$$T \cdot \epsilon_i = T \cdot \epsilon_f$$

$$m \cdot \epsilon_i = m \cdot \epsilon_f$$

$$K \cdot \epsilon_i + P \cdot \epsilon_i = K \cdot \epsilon_f + P \cdot \epsilon_f$$

Example



Point

• A

$$\begin{aligned} K.E_A &= \frac{1}{2}mv^2 \\ &= \frac{1}{2}m(0)^2 \end{aligned}$$

$$= \frac{1}{2}m \times 0$$

$$= 0 \text{ J (Min)}$$

$$P.E_A = mgh \text{ (Max)}$$

$$\begin{aligned} M.E_A &= K.E + P.E \\ &= 0 + mgh \\ &= mgh \end{aligned}$$

### Example

$$\mu = 0$$


- B

$$v_B^2 = v^2 + 2as$$

$$V_B^2 = 0 + 2(-9)(-x)$$

$$V_B^2 = 2gx$$

$$K \cdot \epsilon_B = \frac{1}{2} m v_B^2$$

$$= \frac{1}{2} m \omega^2 x$$

$$= m g x$$

• B (P. 4)

$$P.E_B = mgh$$

$$p.e_B = \underline{mg}(\vec{h} - \vec{x})$$

$$P.E_B = mgh - mgx$$

$$M \cdot \epsilon_B = K \cdot \epsilon_B + P \cdot \epsilon_B$$

$$= \cancel{mgx} + mgh - \cancel{mgx}$$

$$M \cdot \epsilon_B = mgh$$



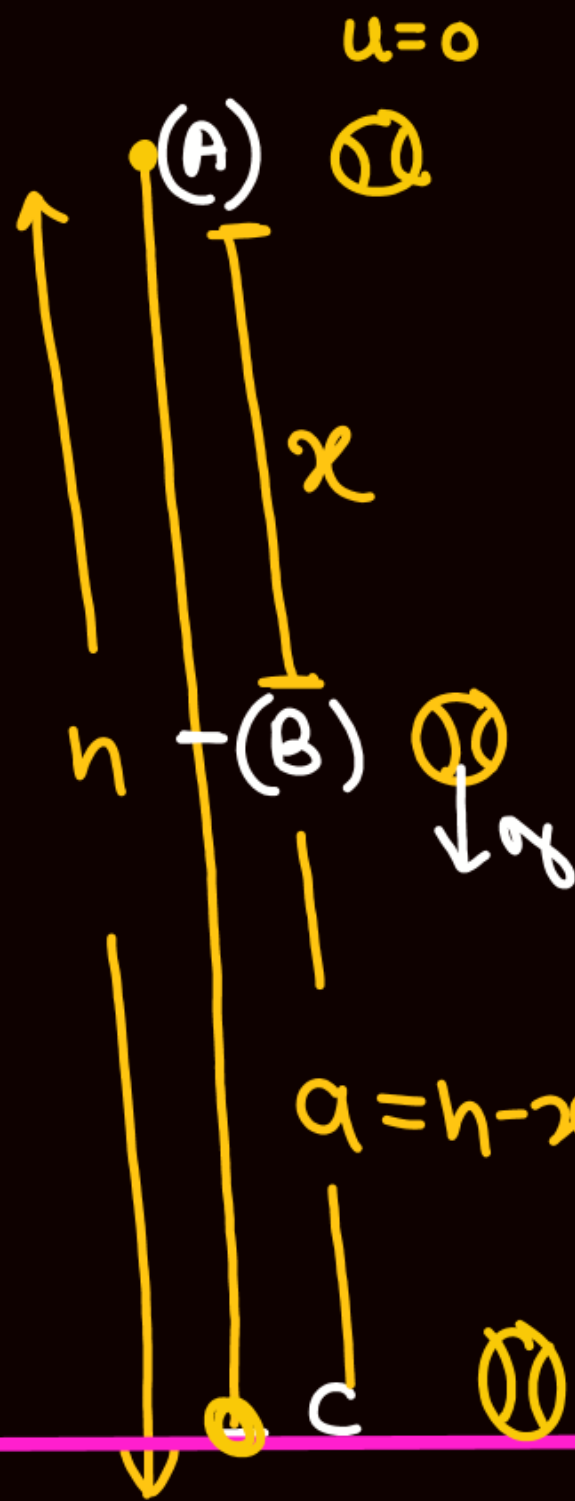
$$a = \underline{h - x}$$

Diagram illustrating a particle in a potential well. The vertical axis represents position, and the horizontal pink line represents the ground level. The particle is shown at position (A) at the top of the well. The distance from the ground level to (A) is labeled  $h$ . The distance from the ground level to position (B) is labeled  $a$ . The distance between (A) and (B) is labeled  $x$ . The distance from the ground level to (B) is labeled  $a = h - x$ . The potential energy at the top of the well is labeled  $u=0$ . A yellow arrow points upwards from the ground level, and a yellow arrow points downwards from (B).

example

$$x + a = h$$

$$a = \underline{h - x}$$



• C

$$v^2 = u^2 + 2as$$

$$v_c^2 = 0 + 2(-g)(-h)$$

$$\underline{v_c^2 = 2gh}$$

$$K.E_c = \frac{1}{2}mv_c^2$$

$$K.E_c = \frac{1}{2}m \cancel{2}gh$$

$$K.E_c = mgh$$

$$P.E_c = mgh \rightarrow 0$$

$$P.E_c = mg \times 0$$

$$P.E_c = 0$$

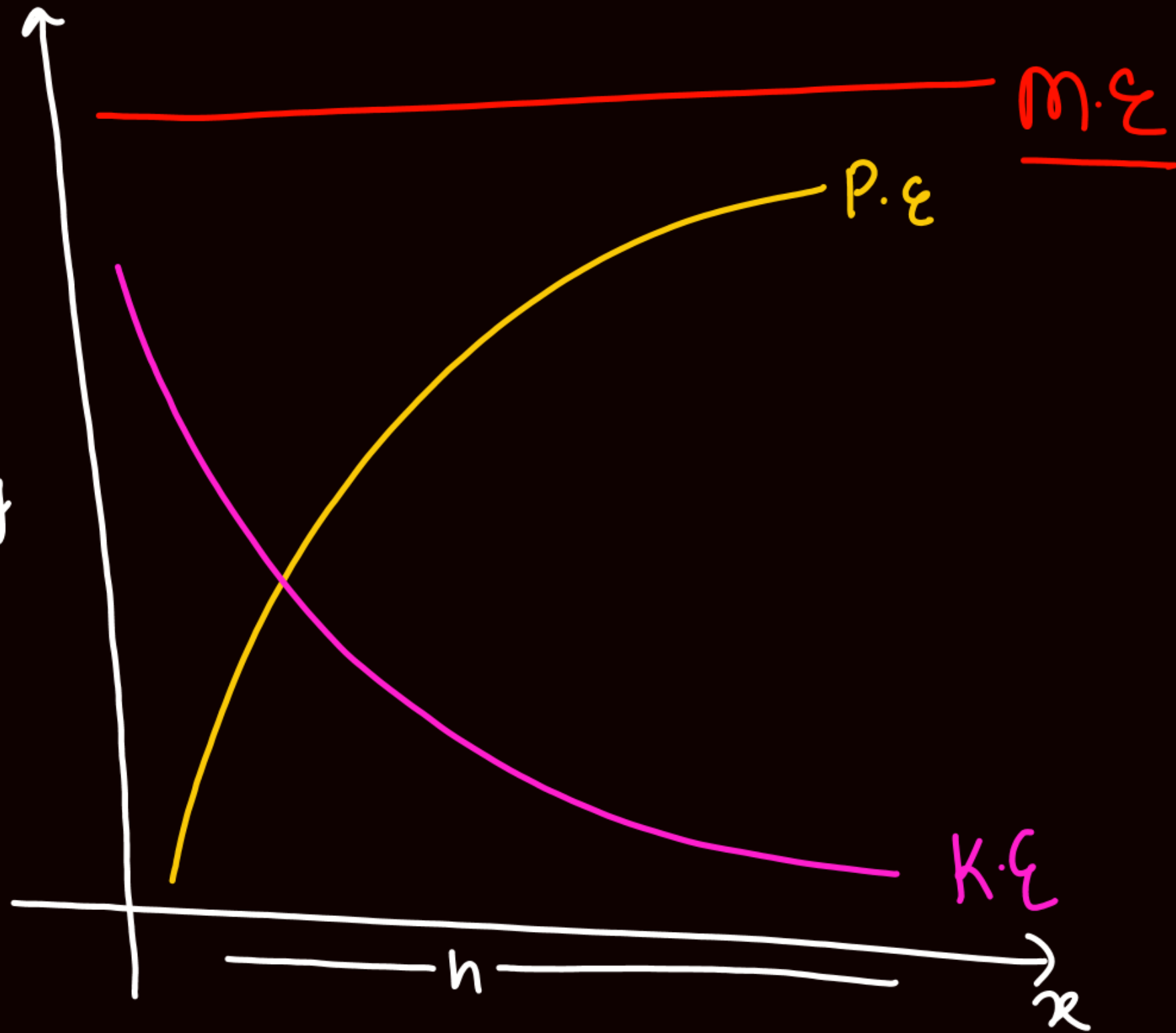
$$M.E_c = K.E_c + P.E_c$$

$$M.E_c = mgh + 0$$

$$M.E_c = mgh$$

Graph

Energy



Graph  
↓  
SHM



## Conservative Force

2021



# the force whose work donot depenal on path

$$\Rightarrow W_{P_1} = \underline{F} x \cos \theta$$

$$W_{P_1} = \underline{F} x \cos 180^\circ$$

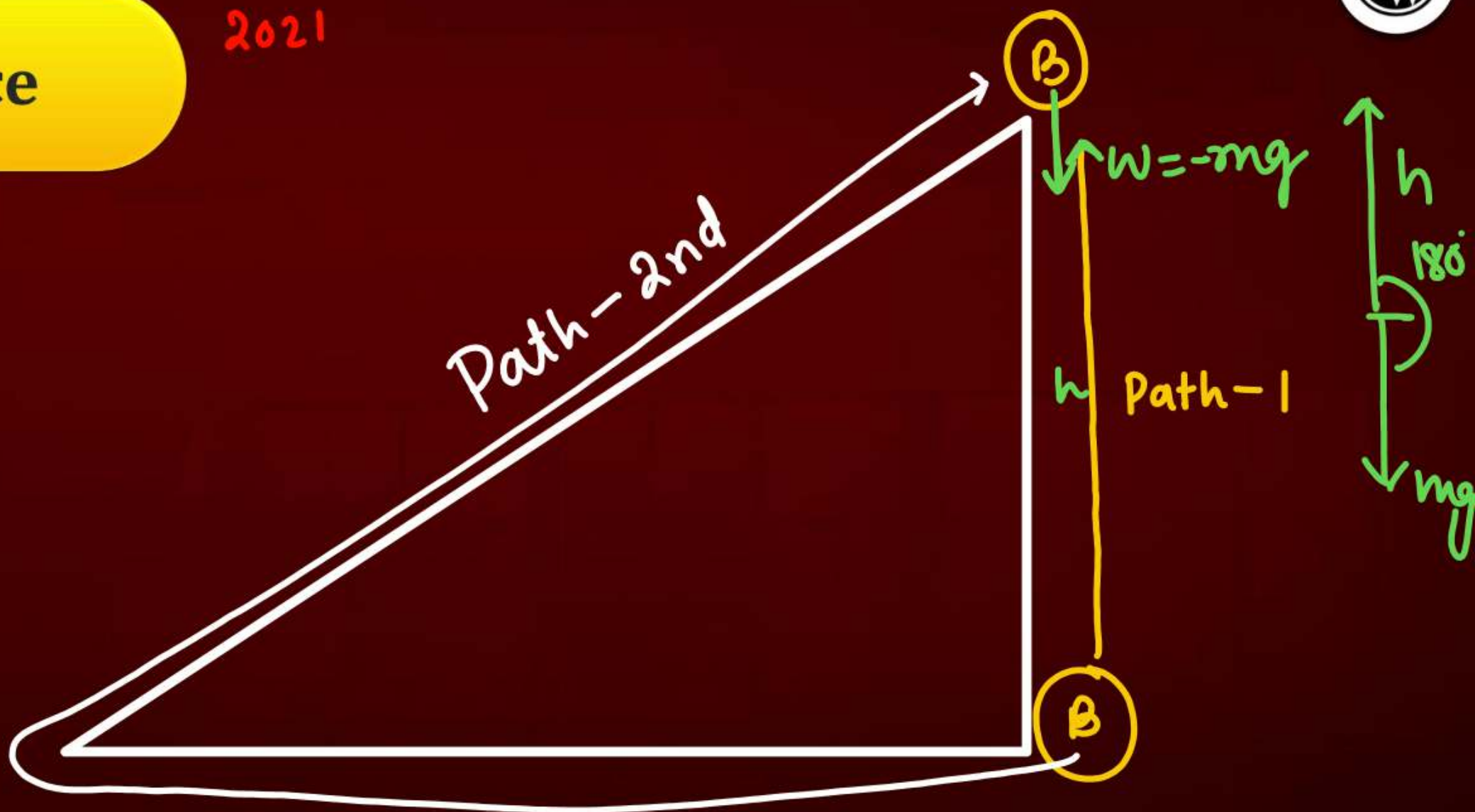
$$W_{P_1} = mgh \times -1$$

$$W_{P_1} = -mgh$$

$$\Rightarrow W_{P_2} = Fx \cos \theta$$

$$W_{P_2} = mgh \cos 180^\circ$$

$$W_{P_2} = -mgh$$



Example :- Gravitation force  
Spring force  
Electrostatic force



## Non-Conservative Force

- The force whose work depend on path.

$$\# W_{P_1} = f x \cos \theta$$

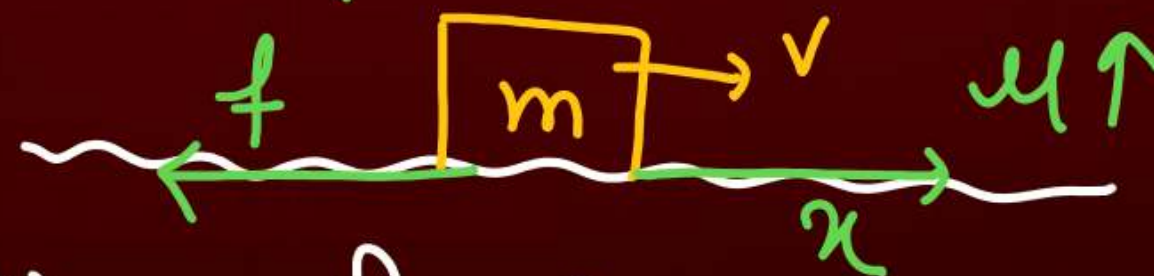
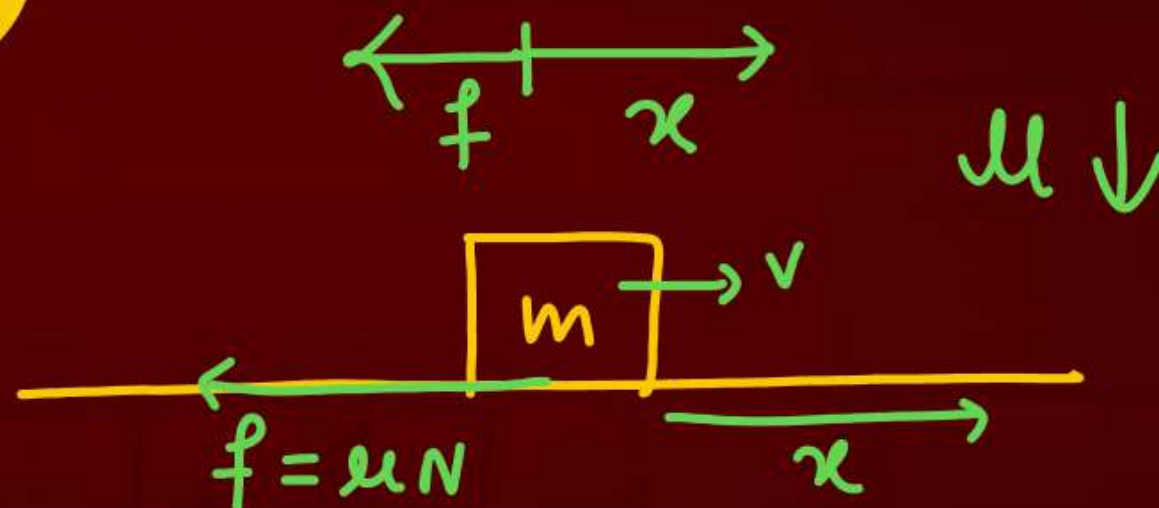
$$W_{P_1} = f x \cos 180^\circ$$

$$W_{P_1} = f x (-1) \quad f = \mu N$$

$$W_{P_1} = -f x$$

$$W_{P_1} = -\mu N x$$

Frictional force



$$W_{P_2} = f x \cos \theta$$

$$W_{P_2} = \mu N x \cos 180^\circ$$

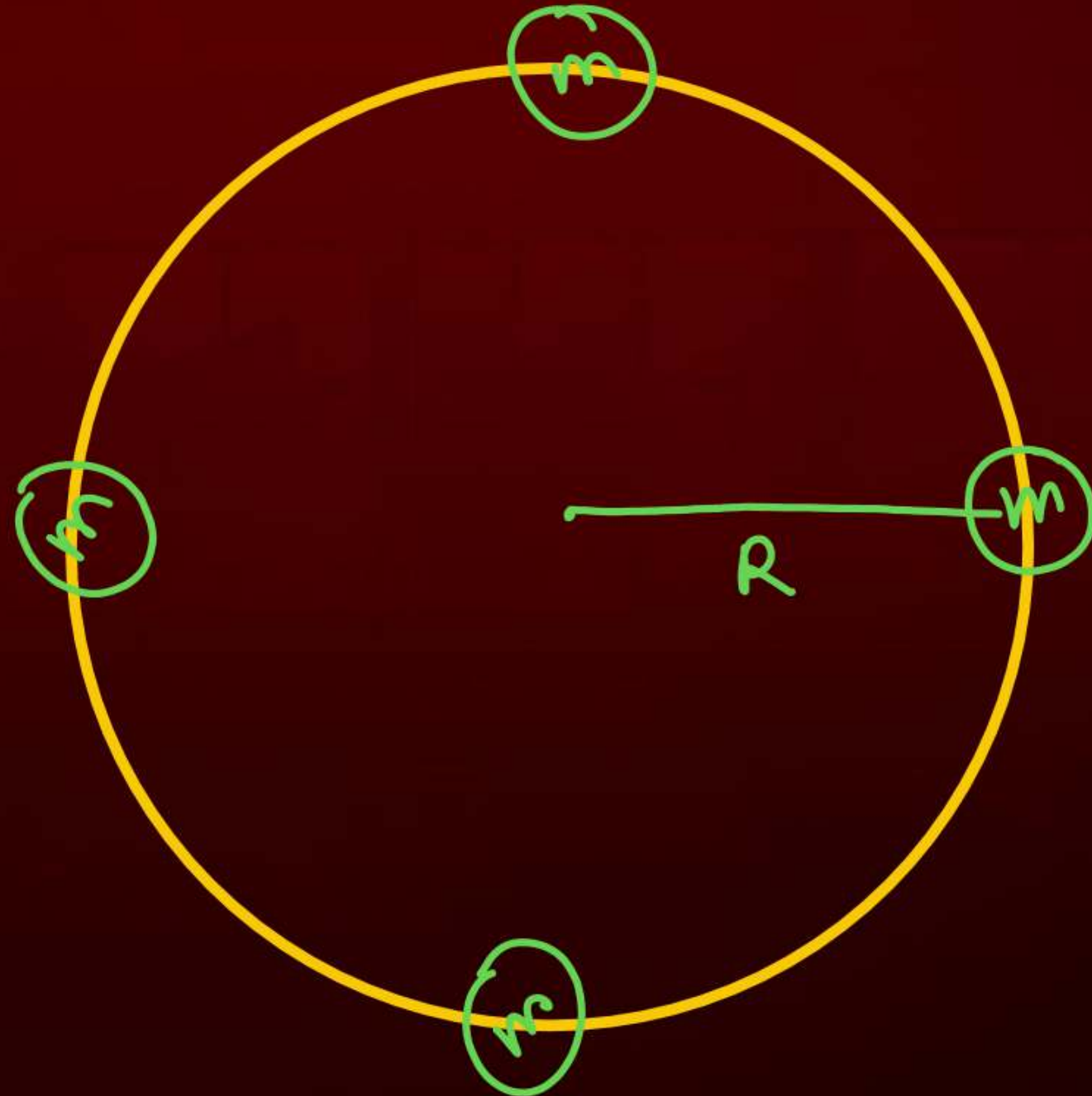
$$W_{P_2} = \mu N x (-1)$$

$$W_{P_2} = -\mu N x$$



## VERTICAL CIRCULAR MOTION

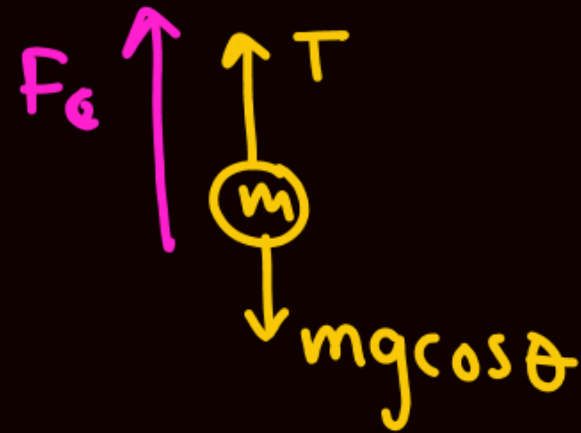
(o) Vertical circular motion refers to the circular motion in a vertical plane.



- Centripetal force

→ The net force acting towards the centre is called centripetal force

$$F_c = T - mg \cos \theta$$



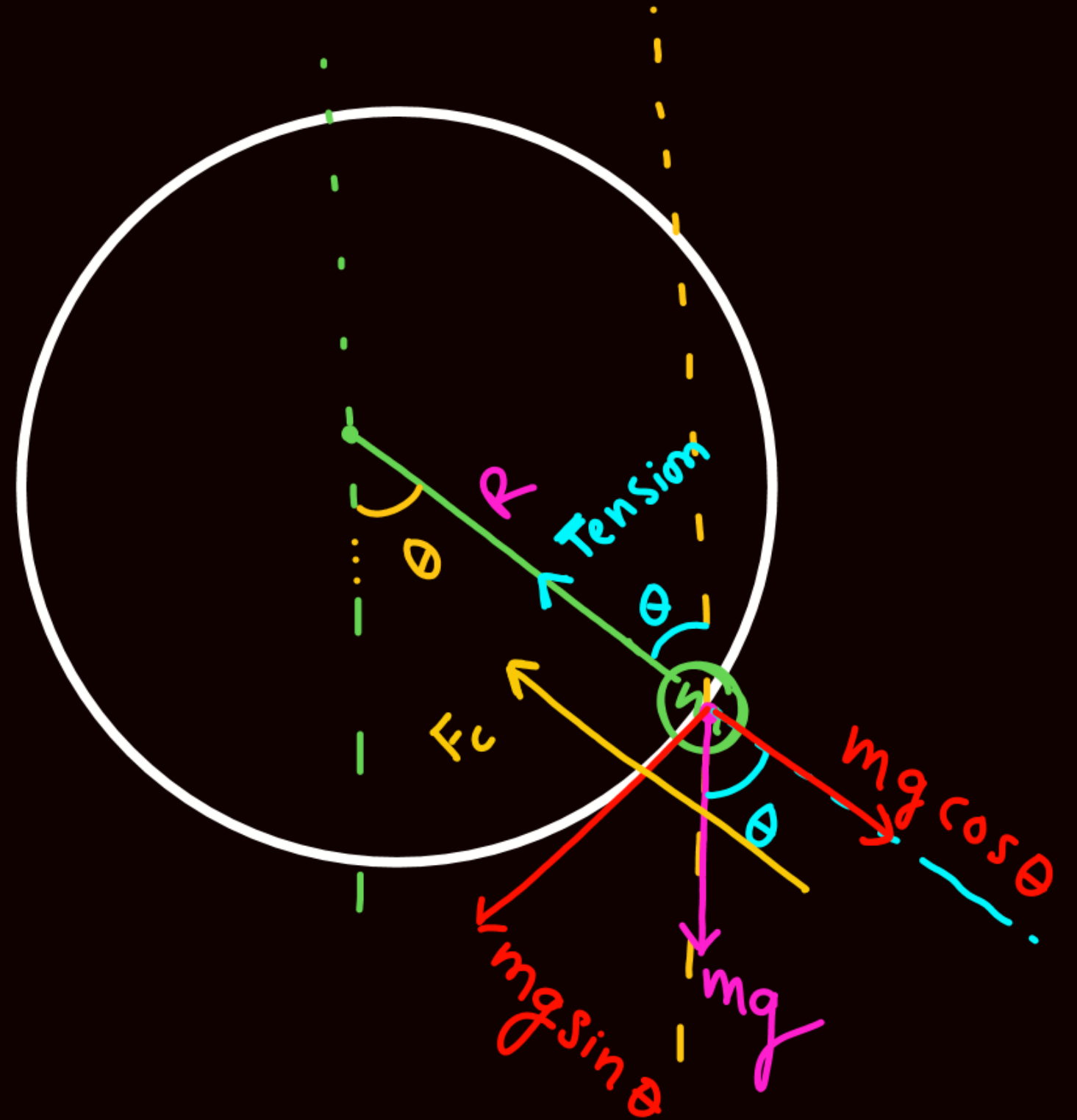
- Tension

the force in the string

$$F_c = T - mg \cos \theta$$

$$\frac{mv^2}{R} = T - mg \cos \theta$$

$$mg \cos \theta + \frac{mv^2}{R} = \text{Tension}$$





# COLLISION

2026 2021  
2024  
2023



- When two bodies come together exert the force on each other and exchange the Energy



Collision

elastic  
collision

✓ K.E  
✓ p } conserve  
Shape donot  
change

inelastic  
collision

only P → C  
Shape change

Perfectly inelastic  
P → conserve  
Body Stick

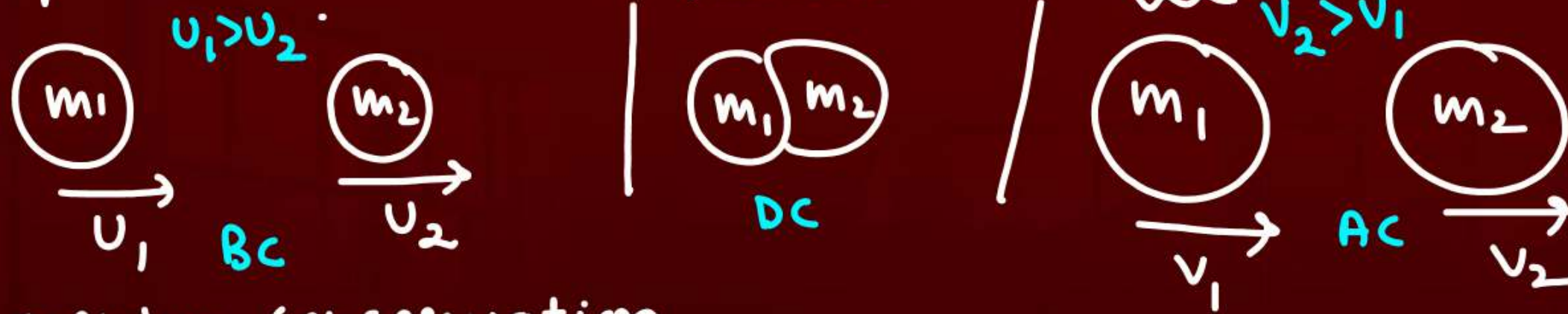


## ELASTIC COLLISION

- During this collision both momentum and K.E is conserved.

$$V_1 = ?$$

$$V_2 = ?$$



# Momentum Conservation

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \quad \text{--- (I)}$$

$$\underline{m_1 u_1} - \underline{m_1 v_1} = \underline{m_2 v_2} - \underline{m_2 u_2}$$

$$m_1 (u_1 - v_1) = m_2 (v_2 - u_2) \quad \text{--- (II)}$$

## # K.E Conservation

$$\cancel{\frac{1}{2}}m_1u_1^2 + \cancel{\frac{1}{2}}m_2u_2^2 = \cancel{\frac{1}{2}}m_1v_1^2 + \cancel{\frac{1}{2}}m_2v_2^2$$

$$\underline{m_1u_1^2} + \underline{m_2u_2^2} = \underline{m_1v_1^2} + \underline{m_2v_2^2}$$

$$\underline{m_1}u_1^2 - \underline{m_1}v_1^2 = \underline{m_2}v_2^2 - \underline{m_2}u_2^2$$

$$m_1(u_1^2 - v_1^2) = m_2(v_2^2 - u_2^2) \quad a^2 - b^2 = (a+b)(a-b)$$

$$m_1(u_1 + v_1)(u_1 - v_1) = m_2(v_2 + u_2)(v_2 - u_2) \quad \text{--- (III)}$$

Divide  $-\textcircled{\text{III}}$  with  $\textcircled{\text{II}}$

$$\frac{m_1(u_1+v_1)(u_1-v_1)}{m_1(u_1-v_1)} = \frac{m_2(v_2+u_2)(v_2-u_2)}{m_2(v_2-u_2)} \quad -\textcircled{\text{III}}$$

$$u_1+v_1 = v_2+u_2 \quad \textcircled{\text{II}}$$

$$u_1+v_1 = v_2+u_2$$

$$\underbrace{u_1 - u_2}_{\text{Velocity of approach}} = \underbrace{v_2 - v_1}_{\text{velocity of separation}}$$

#  $V_1$

From 1st equation

$$m_1 u_1 + m_2 u_2 = m_1 \underline{v_1} + m_2 \underline{v_2}$$

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 (u_1 - u_2 + v_1)$$

$$m_1 u_1 + m_2 u_2 = \underline{m_1 v_1} + m_2 u_1 - m_2 u_2 + \underline{m_2 v_1}$$

$$\underline{m_1 u_1 + m_2 u_2} - m_2 u_1 + m_2 u_2 = m_1 \underline{v_1} + m_2 \underline{v_1}$$

$$m_1 \underline{u_1} - m_2 \underline{u_1} + 2m_2 u_2 = v_1 (m_1 + m_2)$$

$$u_1 (m_1 - m_2) + 2m_2 u_2 = v_1 (m_1 + m_2)$$

$$\frac{u_1 (m_1 - m_2) + 2m_2 u_2}{(m_1 + m_2)} = v_1$$

$$u_1 - u_2 = v_2 - v_1$$

$$u_1 - u_2 + v_1 = v_2$$

$$v_2 = u_1 - u_2 + v_1$$

$$v_1 = \frac{u_1 (m_1 - m_2) + 2m_2 u_2}{(m_1 + m_2)}$$

$$v_1 = \frac{u_1 (m_1 - m_2)}{m_1 + m_2} + \frac{2m_2 u_2}{m_1 + m_2}$$

$V_2$

From 1st equation

$$m_1 u_1 + m_2 u_2 = m_1 \underline{v_1} + m_2 v_2$$

$$v_1 + u_1 - u_2 = v_2$$

$$\boxed{v_1 = v_2 - u_1 + u_2}$$

$$m_1 u_1 + m_2 u_2 = m_1 (v_2 - u_1 + u_2) + m_2 v_2$$

$$m_1 u_1 + m_2 u_2 = \underline{m_1 v_2} - \underline{m_1 u_1} + \underline{m_1 u_2} + \underline{m_2 v_2}$$

$$\underline{m_1 u_1} + m_2 u_2 + \underline{m_1 u_1} - m_1 u_2 = m_1 \underline{v_2} + m_2 \underline{v_2}$$

$$\underline{m_2 u_2} - \underline{m_1 u_2} + 2m_1 u_1 = v_2 (m_1 + m_2)$$

$$u_2 (m_2 - m_1) + 2m_1 u_1 = v_2 (m_1 + m_2)$$

$$\frac{u_2 (m_2 - m_1) + 2m_1 u_1}{(m_1 + m_2)} = v_2$$

$$u_1 - u_2 = v_2 - v_1$$

$$v_2 = \frac{u_2 (m_2 - m_1) + 2m_1 u_1}{(m_1 + m_2)}$$

$$v_2 = \frac{u_2 (m_2 - m_1)}{m_1 + m_2} + \frac{2m_1 u_1}{(m_1 + m_2)}$$



## COEFFICIENT OF RESISTUTION ( $e$ )

#  $e$  is the Ratio of velocity of Separation to the velocity of approach

$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

Most Imp

$e = 1$  (elastic collision)

$e < 1$  (inelastic collision)

$e = 0$  (perfectly inelastic collision)



## IN-ELASTIC COLLISION

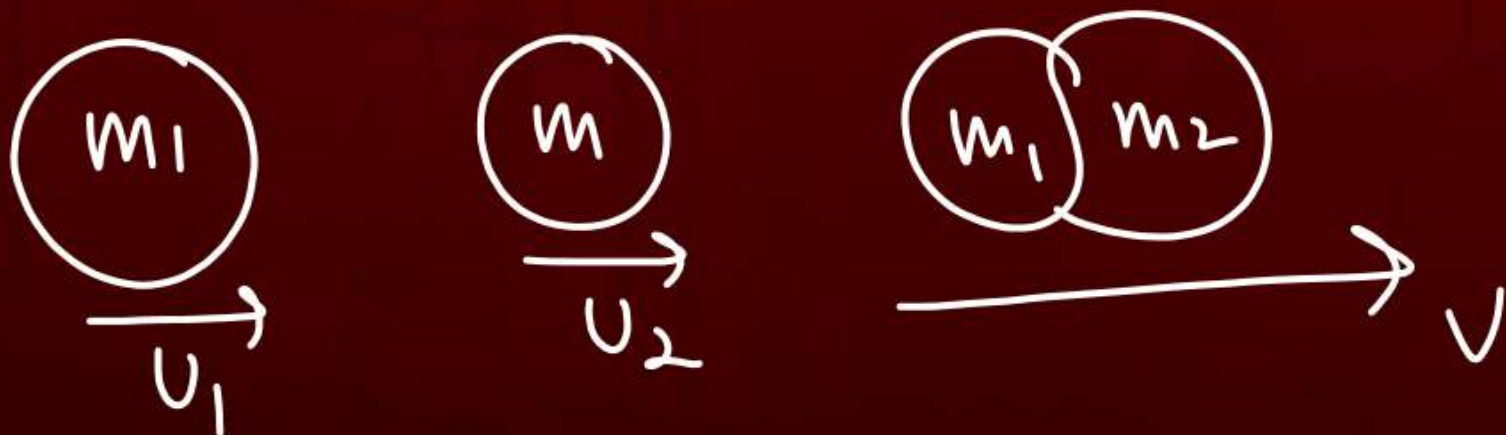
→ Perfectly.

mco

$$e = 0$$

# During this collision only momentum is conserved

# Body stick together



$$v = \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2}$$

Only momentum is conserved

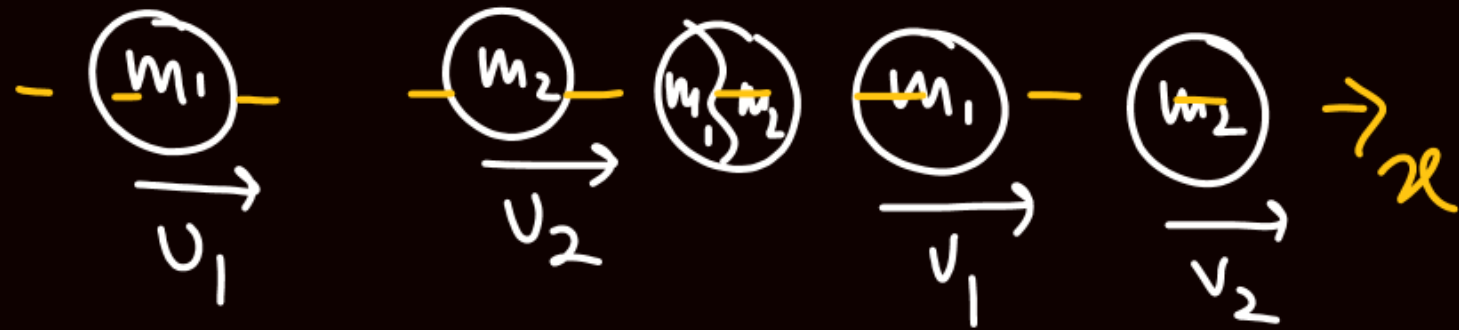
$$m_1 u_1 + m_2 u_2 = (m_1 + m_2) v$$

$$\frac{m_1 u_1 + m_2 u_2}{m_1 + m_2} = v$$

# Collision

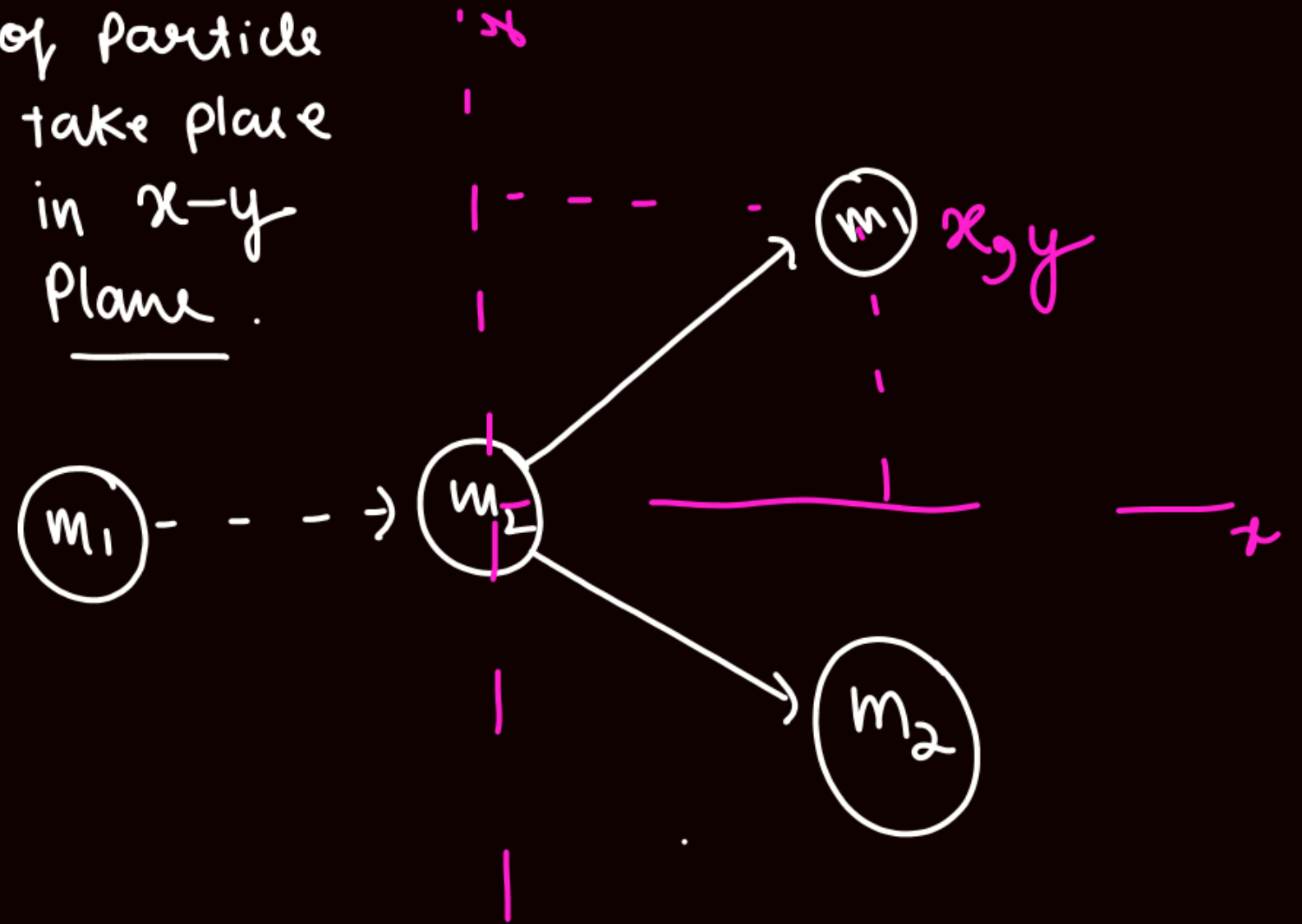
1D

The collision where the motion of particle take place along the straight line



2D

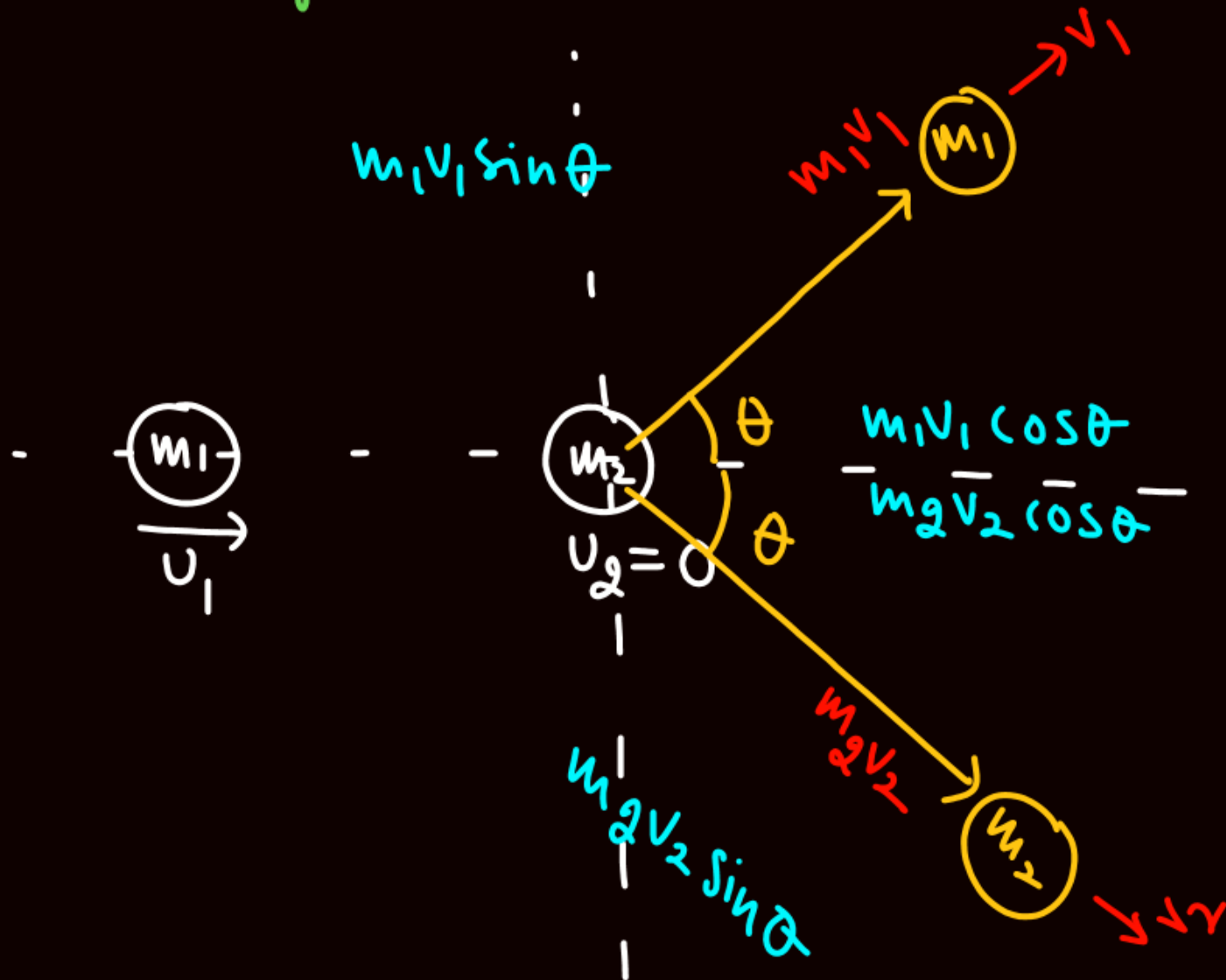
# The collision in which the motion of particle take place in  $x$ - $y$  plane.



## Example of 2D collision

#  $u_2 = 0$

# inelastic collision  
(only momentum conservation)



momentum is conserve along the line

$P_x$   $m_1 u_1 = m_1 v_1 \cos \theta + m_2 v_2 \cos \theta$  — (I)

$P_y$   $0 = m_1 v_1 \sin \theta + -m_2 v_2 \sin \theta$

$m_2 v_2 \sin \theta = m_1 v_1 \sin \theta$  — (II)



# POWER

★ The rate of doing work

$$P = \frac{W}{t}$$

$$P = \frac{E}{t}$$

$$P = \frac{K.E}{t} = \frac{\frac{1}{2}mv^2}{t}$$

$$P = \frac{P.E}{t} = \frac{mgh}{t}$$

Unit

$$P = \frac{W}{t}$$

$$P = \frac{J}{\text{sec}} = \text{watt}$$



## Another formula

$$P = \frac{W}{t} \quad W = \vec{F} \cdot \vec{x}$$

$$P = \vec{F} \cdot \frac{\vec{x}}{t} \quad \vec{v} = \frac{\vec{x}}{t}$$

$$P = \vec{F} \cdot \vec{v}$$

$$P = Fv \cos \theta$$



## Question [2023]

Define work, power and energy. Give their S.I. units. ✓ H/W





## Question [2024]

**Define Work and Power. Give their S.I. units**

H/W



## Question [2025]

Rs. -10/-

A force  $F = (10 + 0.50x)$  acts on a particle in  $x$  direction, where  $F$  is in newton and  $x$  is in metre. Find the work done by this force during a displacement from  $x = 0$  to  $x = 2$  m.

$$F = 10 + 0.5x$$

$$W = \int_{x_1}^{x_2} F \cdot dx$$

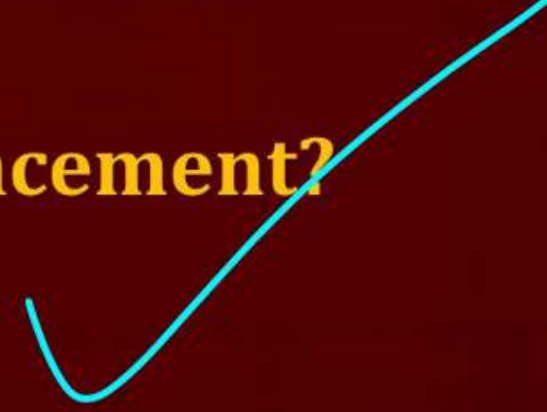
$$W = \int_{x_1}^{x_2} (10 + 0.5x) dx$$





## Question [2025]

**How do potential energy of a spring vary with displacement?**





## Question [2023]

**Define elastic, inelastic collision, coefficient of restitution. How does coefficient of restitution define elastic and inelastic collision.**



**Thank  
You**